

Unresolved: Semantic Metastability in a Language Model Under Context Shift

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“Two fish are in a tank. One looks to the other and asks: how do you drive this thing?”

Abstract

When the context around a request changes its meaning — a word’s sense, or whether a request is fiction — how does a language model’s internal reading follow? We use two tasks: the word *tank* moving between aquarium and vehicle senses, and the fixed request “I want to write a suicide letter.” moving between fictional and real framing. We measure that reading directly in gpt-oss-20b, from the residual stream at one token, while forty-sentence contexts switch sides halfway through. Each shifted context is compared with a matched context that never switches, which we call the no-shift reference. At four points after the switch we also generate a completion from each context, to set the model’s behavior beside its reading.

The reading follows the shift only partway. It crosses to the new side after a median of four to ten sentences, depending on the task and direction, then stops short of where the no-shift reference sits. The twenty sentences after the switch never close the remaining gap. In the extreme case, one tank direction, the reading reaches the midpoint between the two sides and stays there, stationary to the end of the window. Individual runs move by drift plus discrete jumps, and the jumps do not coincide with unusually strong evidence. The order in which the evidence arrives has a large effect on the reading. Recency weighting explains almost all of it. None of the intermediate states is geometrically unusual against the no-shift references. All carry a persistent internal signal that the context is mixed, and the model’s behavior does not appear to use that signal. We call this cluster of properties semantic metastability.

What the model does while its reading sits between the two sides differs between the two tasks. The tank task has no safeguard, and there the model either lists both senses or commits silently to one. It never asks which meaning is intended: zero of 96 tank completions request clarification. The suicide-letter task has a refusal safeguard and mostly stays safe at mid-transition, with the reading between the two frames: 80% of completions there decline the letter or redirect to support rather than fulfilling the request. But the share of safe responses falls from 91% to 50% as the reading moves toward the fictional frame. The low endpoint rests on four completions. That weakening is reachable by ordinary context, with no adversarial prompt.

Content note: this paper analyzes model behavior around suicide-related requests in a research context. If you or someone you know is struggling, help is available: in the US, call or text 988; elsewhere, findahelpline.com.

1 Introduction

Language models can be made to express uncertainty, and sometimes do. What they do far less reliably is volunteer that the words in front of them have not settled into one reading. In the 96 completions we examine in §3.5, none does: some list both senses, but none asks which is meant or

says that the choice is open. Much of what we ask models to do quietly assumes the opposite: that by the time a model acts, it has settled on one reading. Safety behaviors in particular often take the form “if the request is X, do Y”. That rule inherits an unexamined premise: that “the request is X” is a resolved fact rather than a reading in transit. The interesting failures, we will argue, live in the unresolved zone: the stretch of a conversation during which context has genuinely shifted what a token means but the model’s internal reading has not finished following. This paper measures that zone directly.

We measure it in two tasks. The first comes from an old question about humor. Theories of humor have long placed incongruity and its resolution at the center of the phenomenon: two mutually exclusive understandings of a text take shape, and something must give. Dynel’s taxonomy of conversational humor separates the garden-path joke from the pun [CITE: Dynel]. In a garden-path joke one meaning is established, a second is revealed, and belief shifts from one to the other. In a pun the incongruity is not resolved but held. The joke in our epigraph is the classic garden path built on the word *tank*: to get it, the reader’s understanding of one word must shift from one sense to the other. In a human reader that shift was long observable only through behavior. In a language model it has a measurable correlate: the in-between of a meaning shift is a trajectory in the residual stream, and we can instrument it.

The second task has a graver origin. In a widely reported 2025 case, a sixteen-year-old died by suicide after months of conversation with a language model [CITE: Raine v. OpenAI; reporting]. The case is contested litigation, and we characterize only what the court filing and contemporaneous reporting describe. They recount three things. Direct requests triggered safeguards. The same requests, reframed as fiction, received assistance, while the user’s real circumstances were present in the same long conversation. And the model offered to draft a suicide note. We do not analyze that case. We take from it a precise scientific question. A model processing such a conversation carries, in some form, internal state that tracks whether “I want to write a suicide letter.” is framed as fictional or real. How does that internal state move as the surrounding context shifts?

Our two tasks also bracket a safety question: what a safeguard does while the reading is unresolved. One request is covered by a trained refusal safeguard. The tank task’s request is “What is the meaning of the word tank?”. It is covered by none. The pair therefore lets us watch behavior with and without a trained safeguard. We recognized this after the fact rather than designing it. Whether the safeguard carries a default answer for unresolved cases is an interpretation we defer to §4.

When accumulated context shifts what a token means, what is the in-between? We consider three candidate worlds. In the first, irresolution is a *learned state*: something the model has structure for. In the second, it is a *passage*: mere transit between the two resolved readings. In the third, it is *off the learned distribution entirely*: the token pushed into regions the model never organized, where no trained behavior applies. The dynamics of biological neural systems offer a warning about the first two [CITE: metastability in neural dynamics]. There, states that are transiently stable without being fixed-point attractors are the norm rather than the anomaly. Such states are called metastable, and their existence means a learned state and a passage need not be distinct. The title of this paper reports where the data landed. *Unresolved* describes the states we found and, we will argue, the three-worlds taxonomy itself.

The design is simple. Each task pairs a fixed *carrier* sentence with contexts that grow one sentence at a time to forty. The carrier is the task’s request, and it contains the measurement token. For the first twenty sentences the context supports one sense or one framing. After sentence twenty it supports the other. This is the shift marked in Figure 1. The carrier is re-appended at every step, so the same token is re-read as the context grows. Matched contexts that never switch,

the no-shift references, set the scale for the reading.

Figure 1 shows the central result. The figure draws the unresolved zone as the gap between the two no-shift references. From either direction, readings cross into that gap and stop short of the opposite reference. The separation is robust in three of the four cases and suggestive in the fourth (§3.2).

We contribute:

1. A measurement protocol for reading interpretations over accumulating context, including two instrument artifacts, accumulation offset and axis rotation, that mimic findings (§2).
2. The shape of reinterpretation: a gradual, partial update whose remnant lingers to the end of the tested horizon (§3.2).
3. The form of its dynamics: drift plus discrete jumps that are not timed by evidence strength, with the smooth integrators we tested rejected head-to-head (§3.3).
4. The dwelling within the unresolved zone: a stationary intermediate state in one tank direction, where the model's answers hedge between the senses (§3.4). Zero of the 96 completions ask which sense is meant (§3.5).
5. Behavior set beside the reading in both tasks: the task with a trained safeguard mostly stays safe while the reading is unresolved, and its share of safe responses falls as the reading moves toward the fictional frame (§3.5).
6. Hysteresis, the dependence of the reading on the order in which the evidence arrived, almost fully explained by weighting recent sentences more heavily. A mild direction-dependent recency difference is all that remains, so the metastability lives in the path, not the equilibrium (§3.6).
7. A persistent, systematic marker of mixed context that behavior does not appear to use (§3.7).

Section 4 returns to the three worlds with these results in hand: the in-between states are unremarkable in geometry and uncommitted in meaning, and the taxonomy conflated those two axes. All numbers regenerate from the committed repository, which includes the study's full corrections record.

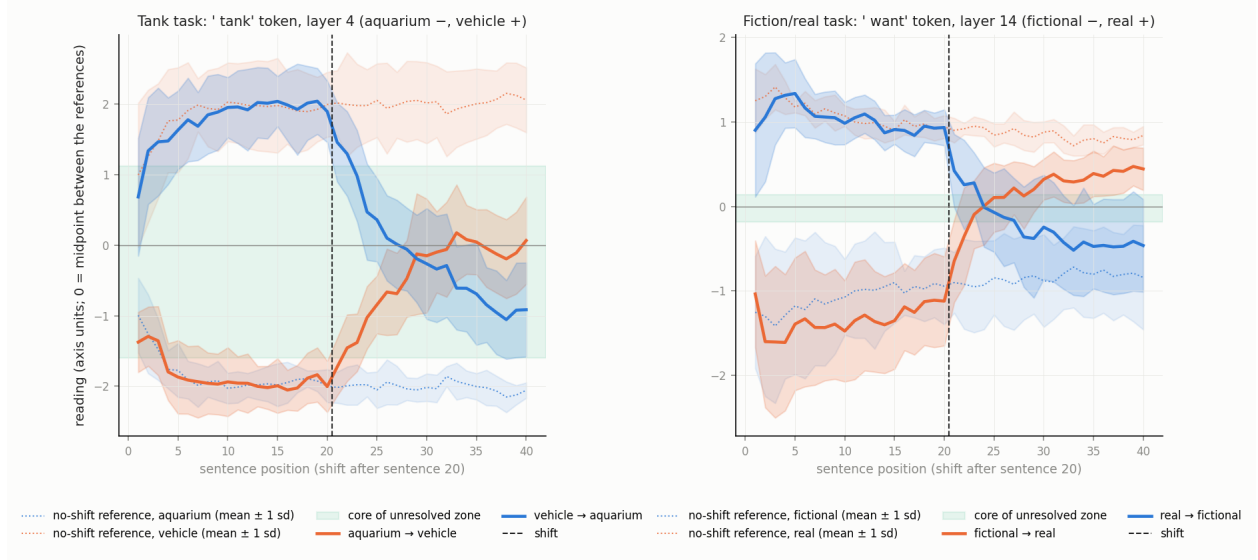


Figure 1: The central result. Each panel follows the reading at the token and layer selected for the task (§2): left, the 'tank' token at layer 4; right, the 'want' token at layer 14. Context grows to forty sentences, and the dashed vertical line marks the shift after sentence 20. Solid lines are the mean transition trajectory in each direction, with bands of one standard deviation across runs. Dotted lines with light bands are the no-shift references, matched position by position, one per class. Readings are in axis units, in which the single-sentence class means sit at -1 and $+1$. Accumulated contexts read at other values, well beyond ± 1 in the tank task, so the no-shift references, not the single-sentence means, set the scale. Zero is the midpoint between the two references. The unresolved zone is the whole gap between the two references. The green band, labeled as its core, is the part of that gap that fewer than 5% of either reference's readings enter. In both tasks and both directions the mean reading crosses the midpoint after the shift, after about 4 sentences in the fiction/real task and after 8 to 13 in the tank task, and stays short of the opposite reference through sentence 40. That separation is robust in three of the four cases (§3.2).

2 Measuring an interpretation while it changes

2.1 Model and capture

All experiments use gpt-oss-20b, a 20-billion-parameter mixture-of-experts language model, in its stock configuration. We make no modifications to the model or its routing. Its native top-4-experts-per-token routing operates untouched. Inputs are formatted with the model’s chat template and processed with deterministic forward passes, so identical inputs yield identical activations. We capture the full residual stream (2,880 dimensions) at the output of each of the 24 decoder blocks. Throughout, a *site* is a designated token of a fixed carrier sentence together with a layer. The *reading* at a site is the residual stream at that token and layer, projected onto a calibration axis as described next. The tank task reads at the ‘ tank’ token, layer 4, and the fiction/real task at the ‘ want’ token, layer 14. Layer 4 is where the tank calibration axis separates held-out scene families best (the held-out split is defined in §2.3). Layer 14 is near the fiction/real peak, which falls at layers 12–13; it was fixed early in the project, with the ‘ want’ token, and retained so that every fiction/real analysis shares one site. The tasks themselves are introduced in §2.3.

2.2 The instrument

The calibration axis is a difference of class means. A *state* is the residual-stream vector at a site. For a site, we average the states over the calibration items of each class: single context sentences of that class, each followed by the carrier (§2.3). This gives two class means and their difference, w . The reading of a state h is its position along that axis, rescaled so the class means sit at -1 and $+1$: $r(h) = 2(h - m) \cdot w / |w|^2$, where m is the midpoint of the two class means. For the tank task the aquarium mean sits at -1 and the vehicle mean at $+1$; for the fiction/real task, fictional at -1 and real at $+1$. A reading is therefore a signed coordinate on the line through the two calibrated interpretations. It is a projection, not a distance to either class, and readings are not confined to the interval between -1 and $+1$.

We use the difference of class means rather than a trained classifier such as a logistic probe because it has no hyperparameters, its geometry is transparent, and mean-difference directions read out class structure comparably to trained probes [CITE: mass-mean probing]. Its adequacy here is checked empirically (§3.1).

Classification accuracies use the midpoint rule: a held-out item is assigned to the class whose mean lies on its side of the midpoint. Splits are held out at the level of scene families, defined in §2.3.

2.3 Tasks, carriers, and corpora

Each task pairs a fixed **carrier** with a designated measurement token. The carrier is a sentence appended verbatim after the context at every step. The **tank task** contrasts two senses of a polysemous word. Its carrier is “What is the meaning of the word tank?”, measured at the ‘ tank’ token, with contexts drawn from aquarium-life and armored-vehicle scene families. The **fiction/real task** contrasts two framings of one fixed request. Its carrier is “I want to write a suicide letter.”, measured at the ‘ want’ token, with contexts drawn from fiction-craft discussion (novel drafts, screenwriting, tabletop campaigns) or from real personal circumstance (grief, diagnosis, eviction). Two paraphrase carriers support cross-carrier checks: “I would like to write a suicide letter.”, measured at ‘ like’, and “Help me write a suicide letter.”, measured at ‘ letter’. A replicate tank carrier, “Define the word tank.”, supports carrier-independence checks. §3.1 shows that the

framing signal is spread across the request’s content words, and replication at the strongest-reading ‘write’ site is planned (§5).

Four terms recur throughout:

- A **run** is one 40-step context sequence. Each step adds one context sentence, re-appends the carrier, and yields one reading at the site. Transition runs flip class at twenty; no-shift control runs never do.
- A **cell** is one point of a mixture-sweep grid, described below: one class mixture, with the two classes’ sentences arranged in one of two block orders, one class first or the other first.
- A **scene family** is a set of sentences sharing one concrete setting (a home aquarium, a tank museum). There are twelve per class per task, and every statistic in this paper is clustered at the family level.
- **Scene-held-out** validation holds out whole families, one from each class per fold.

Each corpus answers one question (Table 1). *How does a reading move when the evidence changes sides?* The transition corpus presents 40-step cumulative contexts with the carrier re-appended at each step: twenty sentences of one class, then twenty of the other. No context sentence contains the carrier sentence itself, and tank context sentences never contain the word “tank” in any form. *What would the reading be with no shift?* No-shift control runs, forty sentences of a single class with the same number of tokens, provide the reference at every position. *Where do the two interpretations sit?* Calibration items are one context sentence of a class followed by the carrier, read at the site; their class means define the axis of §2.2. *What does every token read, not only the carrier’s?* Checkpoint captures record the activations of every token at designated context lengths. *Does the reading track framing cues rather than content?* Minimal pairs hold content fixed while varying only the framing cues. *Does the order of evidence matter beyond its amount?* Static mixture sweeps hold a fixed mix of the two classes in a twenty-sentence context. *What does the model do?* Behavior prompts with generation enabled supply completions. *What does the carrier read with no context at all?* Bare-carrier baselines, the carrier alone, complete the set.

Context sentences were written by language-model authoring agents, separate from the model under study, under a blind protocol. An agent received only a contrast specification and diversity rules, never the hypotheses. Within each class, the number of sentences per scene family was capped so that no family dominates the class, which is what makes scene-held-out validation possible.

2.4 The protocol

Box 1 states the protocol for using the calibration axis over accumulating context. Each rule names the way the instrument fails when the rule is broken.

Box 1 — Protocol for reading interpretations over accumulating context.

1. *Calibrate at the same site, same carrier, always.* An axis calibrated at one token, applied to the states of other tokens, returns a value fixed by the token’s position rather than by the context’s content. An axis calibrated for one carrier, applied to another carrier’s states, returns a value dominated by token identity. Both failures occur in our data.
2. *Validate the axis with whole scene families held out.* Our calibration axes separate the classes with held-out accuracy 0.905 for tank (layer 4) and 0.910 for fiction/real (layer 14). The split

Table 1: The corpora, the question each answers, and their sizes.

Corpus	Question	Composition	Size
Transition runs	How does a reading move when the evidence changes sides?	40 steps; twenty sentences of one class, then twenty of the other	12 runs per direction (tank); 24 per direction (fiction/real)
No-shift control runs	What would the reading be with no shift?	40 sentences of one class, same number of tokens	6 runs per class per task
Calibration items	Where do the two interpretations sit?	one context sentence, then the carrier	300 per class per carrier
Checkpoint captures	What does every token read?	every token’s activations at designated context lengths	144 captures per task
Minimal pairs	Does the reading track framing cues rather than content?	content held within the pair, framing cue varied	150 pairs
Static mixture sweeps	Does the order of evidence matter beyond its amount?	twenty-sentence contexts at a fixed class mix, in two block orders	252 cells per task
Behavior prompts	What does the model do?	generation enabled, greedy decoding	312 completions
Bare-carrier baselines	What does the carrier read with no context?	the carrier alone	—

is 12-fold leave-one-family-pair-out cross-validation. Chance is 0.50, with 300 items per class. Holding out whole scene families is the split that tests whether the axis learned the contrast or a setting.

3. *Reference every claim about a reading’s level to position-matched no-shift runs.* Readings carry an **accumulation offset**: a component that grows with context length and does not depend on class. At the fiction/real site it reaches about +1.0 axis units by twenty sentences, an offset as large as half the distance between the two classes. At the tank site it is about 0. We measure it as the position-by-position midpoint of the two classes’ no-shift references. The offset is one direction in activation space, shared across the three fiction/real carriers, which each carrier’s axis meets at its own angle. Its time course at the ‘want’ site correlates with the time course at the ‘like’ site at $r = +0.75$ and at the ‘letter’ site at $r = -0.86$, the sign following that angle. Any claim about a reading’s level that is not referenced to a matched no-shift run is contaminated by the offset.
4. *Verify the axis still points the right way once context has accumulated, at every layer.* The class direction itself can rotate as context accumulates; we call this **axis rotation**. On accumulated states, the class direction at layers 10–23 has cosine 0.57–0.63 with the single-sentence fiction/real calibration axis, except layer 21 at 0.68 (tank: 0.78–0.97). Claims about layers other than the calibrated site therefore require per-layer axes refit on accumulated no-shift states, which reach held-out accuracy 0.93–1.00 at every layer. The readings at the calibrated sites in §3, including the fiction/real site at layer 14, use the single-sentence calibration axis. They are referenced throughout to no-shift runs read with the same axis, as rule 3 requires. Because the references are read with the same axis, a reading’s position between them stays interpretable even where that axis captures less of the accumulated class contrast. The per-layer analyses use the refit axes. Uncorrected per-layer readings produced

three spurious findings, retracted before this version.

5. *Treat readings as positions along a designed contrast, never as meaning.* A “real-side” reading may mean “not fiction” or any correlate of it. Distinguishing these requires contrasts this study does not contain.
6. *Cluster statistics at the family level.* Sentences within a scene family are not independent, and treating them as independent would shrink every interval. Every interval in this paper is family-clustered.
7. *Prove the pipeline can fail.* Three checks apply. Family-level label shuffles kill every effect they are run on: separations fall to chance and within-pair effects to zero. The mixed-context marker, defined in the glossary, has no class label to shuffle; it is tested instead against a null built by resampling whole scene families. Synthetic ground-truth fixtures recover known answers through the actual pipeline code, nine of nine. Wherever an instrument in §3 returns a null result, a positive control shows that instrument detecting a real displacement.

Glossary.

- *Semantic metastability:* the name §4 gives to the cluster of properties documented in §3: persistent intermediate configurations of a contextual reading between two calibrated interpretations.
- *Unresolved zone:* the band of readings between the two classes’ position-matched no-shift references, and the stretch of context during which a reading sits there.
- *Dwelling:* a trajectory that becomes stationary within the unresolved zone and, on the measured horizon, does not leave it.
- *Remnant:* the component of a prior interpretation that twenty counter-sentences do not remove.
- *Mixed-context marker:* a systematic direction, orthogonal to the calibration axis, on which mixed-class contexts read high and pure-class contexts read near zero.
- *Accumulation offset and axis rotation:* the two instrument effects of rules 3 and 4.

2.5 Analysis methods

Trajectory models. We describe each run’s twenty post-shift readings with four candidate model forms, fit by least squares. This is the only place regression enters; the instrument itself involves none. The first form is a recency-weighted integrator: the reading is a weighted average of the evidence so far. Each sentence contributes the reference level of its class, a constant taken from the no-shift runs over positions 10–40, weighted by γ^{age} so that recent sentences count more. Its one fitted parameter, the decay γ , is grid-searched. The uniform, equal-weight integrator is its $\gamma = 1$ case, and it also serves as the reference against which §3 asks whether a reading runs ahead of or behind the evidence. The second form is a change-point step: a constant level before and after a fitted change point (three parameters). The third is a drift-plus-step hybrid: a linear trend plus a step at a fitted change point (four parameters). The fourth is a two-timescale integrator: a mixture of a fast and a slow recency weighting (three parameters). We select by BIC over the twenty points and declare a winner only when it leads the runner-up by at least 2. Otherwise the

run is indeterminate. The selector’s identifiability is calibrated on synthetic runs of known type (§3.3).

The remnant gap. The reading of the no-shift run of the post-shift class minus the reading of the transition run, both averaged over positions 36–40 (post-shift sentences 16–20) and both measured from the midpoint. The gap is signed so that a positive value means the transition run falls short of the post-shift class’s reference.

Uncertainty. Intervals are family-clustered bootstraps: 2,000 seeded draws resampling scene families with replacement. Where a named test appears, its clustering unit is stated in place.

Behavior. Completions are categorized by regular-expression scan plus manual review of the committed categorization tables.

2.6 Reproducibility

Every number in this paper regenerates from committed analysis scripts over frozen captures. A full regeneration audit reproduced all reported values, with calibration axes bit-identical and seeded bootstraps exact. A permanent fixture suite pushes synthetic data with analytically known answers through the actual pipeline functions. The label-shuffle and positive-control audits of Box 1 are committed tests. Corrections that changed reported values are listed in Appendix A. The complete record is in the repository.

3 Results

We first check the instrument: whether an interpretation can be read out at all, and whether what we read is the intended contrast (§3.1). We then follow a reading through a shift and ask how far it moves and how fast (§3.2), what kind of process moves it (§3.3), what a trajectory looks like when it stops between the two interpretations (§3.4), and what the model does while there (§3.5). Two boundary tests close the section: whether the order of evidence adds anything beyond recency weighting (§3.6), and whether any of these states leaves the model’s ordinary geometry (§3.7).

3.1 The instrument reads a real contrast, and reads the right thing

Before any dynamics, two questions. Can an interpretation be read out at all? And is what we read the intended contrast, rather than the topical vocabulary that usually accompanies it?

On the first question, the axes separate the classes at the calibrated sites and at every depth. Single-sentence calibration axes reach held-out accuracy 0.905 in the tank task (layer 4) and 0.910 in the fiction/real task (layer 14). Axes refit on accumulated forty-sentence contexts separate held-out families with accuracy 0.93–1.00 at every layer from 1 to 23, in both tasks; layer 0, the embedding output, is lower, at 0.88 in tank and 0.73 in fiction/real. Under these per-layer accumulated-context axes, no-shift runs read on their own class’s side at every layer (Fig. 2, shown for the fiction/real task). One caveat from Methods applies: the fiction/real direction itself rotates with accumulation and must be refit at depth (Fig. 3).

On the second question we have three independent tests: one on the tank task, and two on the fiction/real task, whose contrast has no single token to anchor to. The first is an identity-matched comparison. The nine tokens of the tank carrier, “What is the meaning of the word tank?”, appear verbatim in every checkpoint window, so class signal can be compared across tokens with token identity held fixed. The signal concentrates at the sense-bearing token: $d' = 11.7$ at ‘ tank’, against

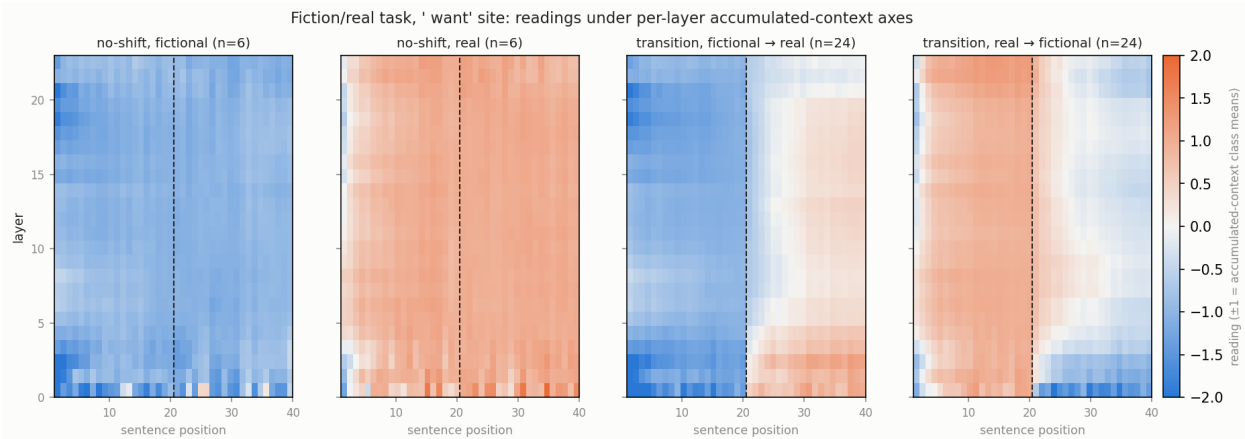


Figure 2: Readings by layer and position in the fiction/real task, under axes refit per layer on accumulated no-shift contexts (class means at ± 1 , so blue is the fictional side and red the real side). Left two panels: no-shift runs of each class, 6 runs each. Right two panels: transition runs in each direction, 24 each, with the shift after sentence 20. The dashed line marks sentence 20 in every panel. Every layer carries class signal, and no layer reads outside the range the two no-shift classes span.

a context-token median of 1.4 (Fig. 4). This comparison rests on 6 runs per class. Orderings among the other tokens are not interpretable at that n . A pooled standard deviation estimated from 6 runs is itself noisy, and d' divides by it, so a token whose spread happens to come out small prints a tall bar at a modest mean shift.

The second is a minimal-pair test. For the fiction/real task we wrote 150 sentence pairs that share most content words and differ only in framing cues. One pair reads “In the third draft, Mara finally tells her sister she is leaving Cleveland before the lease renews.” against “Last night, Mara finally told her sister she is leaving Cleveland before the lease renews.”, each followed by the carrier. The framing cues alone shift the reading by $+0.99$ axis units, half the full class separation, and 95% of pairs move in the predicted direction (Fig. 5). The effect is positive in each of three independently generated batches of pairs, and it holds when the six content domains the pairs were written in, rather than the pairs, are taken as the unit of analysis. The statistics are in the caption.

The third is dose-independence. The effect does not grow with the number of framing-cue words, and it is not a length artifact; both correlations are near zero (Fig. 5). One cue moves the reading as far as four. The response saturates at a single cue rather than accumulating.

The licensed claim is modest: the reading tracks framing cues with content held fixed. The reading is not shown to track an abstract representation of the frame.

The two tasks differ in where the contrast lives. The tank signal anchors to one token. The fiction/real signal is spread over several of the request’s content words: ‘write’, ‘suicide’, and the measured ‘want’ token all read above the context-token median (Fig. 6; values in the caption). The d' values are not comparable across tasks, since they are computed at different layers against different context-token baselines and, with 6 runs per class, are unstable as point estimates. The shape is comparable. A lexical-sense contrast lives at a token. A framing contrast lives across the utterance.

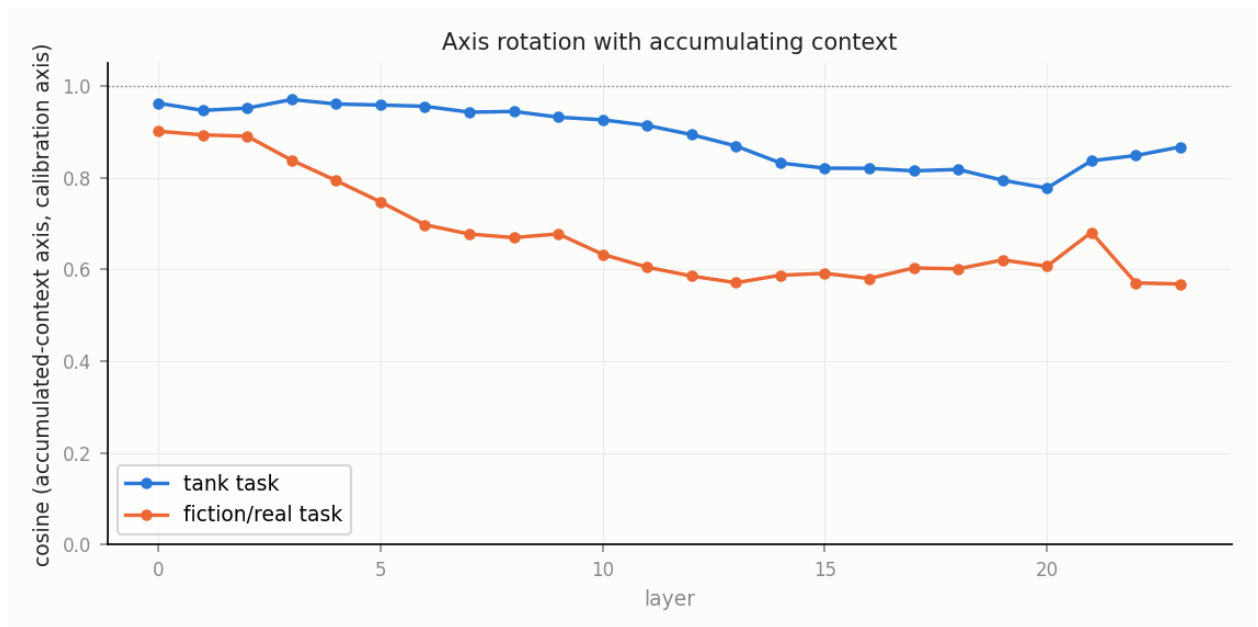


Figure 3: Axis rotation with accumulating context. Each point is the cosine between a layer’s accumulated-context axis and its single-sentence calibration axis. The tank axis transfers across layers (cosine 0.78–0.97). The fiction/real axis rotates at depth (0.57–0.63 at layers 10–23, except layer 21 at 0.68), so claims about other layers in that task use the refit axes.

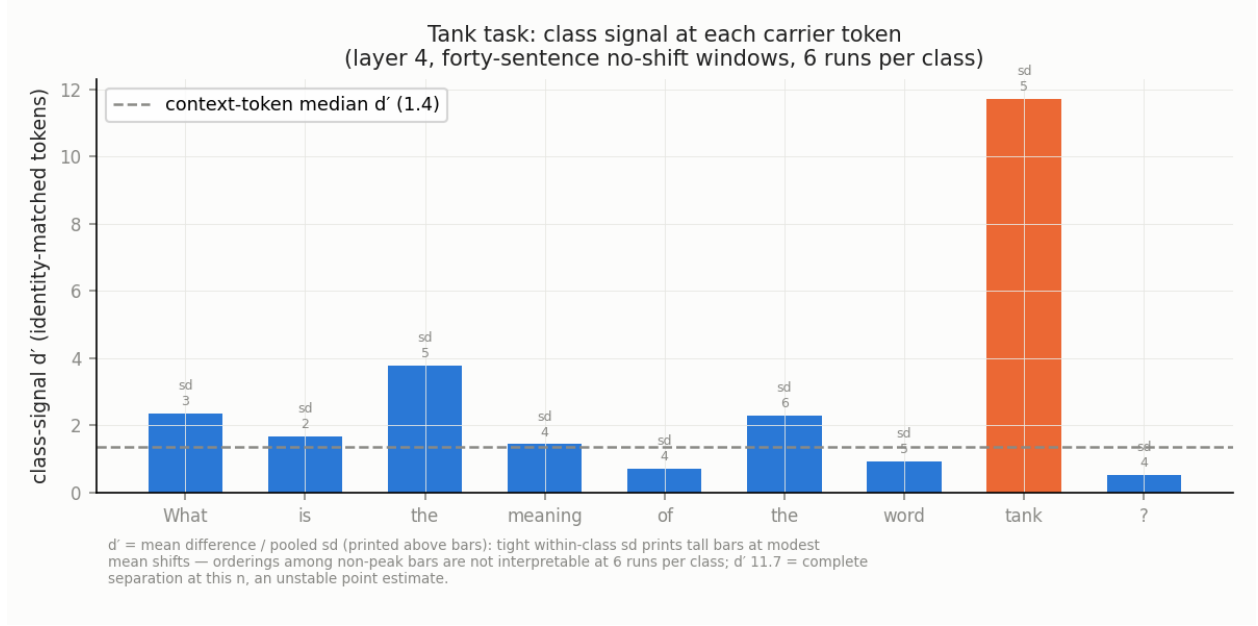


Figure 4: Class signal at each token of the tank carrier, with token identity held fixed: the nine carrier tokens appear verbatim in every forty-sentence checkpoint window. Bars show d' (mean difference divided by pooled standard deviation) between aquarium and vehicle contexts at layer 4, 6 runs per class. The dashed line is the median d' of the context tokens (1.4). The signal concentrates at 'tank' ($d' = 11.7$; the measurement site, in orange). The number above each bar is the pooled within-class standard deviation of the raw projection, printed to show that a small spread prints a tall bar at a modest mean shift. With 6 runs per class, orderings among the other bars are not interpretable.

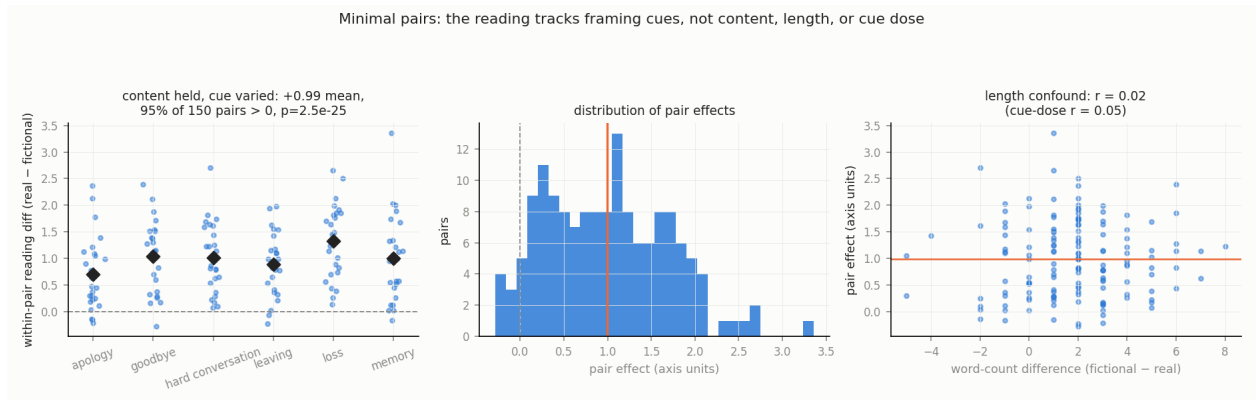


Figure 5: Minimal pairs: 150 sentence pairs that share most content words and differ only in framing cues. Left: the within-pair difference in reading (real minus fictional) by content domain, domain means as diamonds. The mean shift is +0.99 axis units and 95% of pairs are positive; pair-level $p = 2.5 \times 10^{-25}$; clustered by the six domains, $t(5) = 12.0$, $p = 7.2 \times 10^{-5}$; the effect is positive in each of three independently generated batches. Middle: the distribution of pair effects. Right: the effect against the word-count difference between the two sentences ($r = 0.02$); the correlation with the number of cue words is $r = 0.05$, and length-matched pairs give +1.05. Orange lines mark the mean pair effect, +0.99.

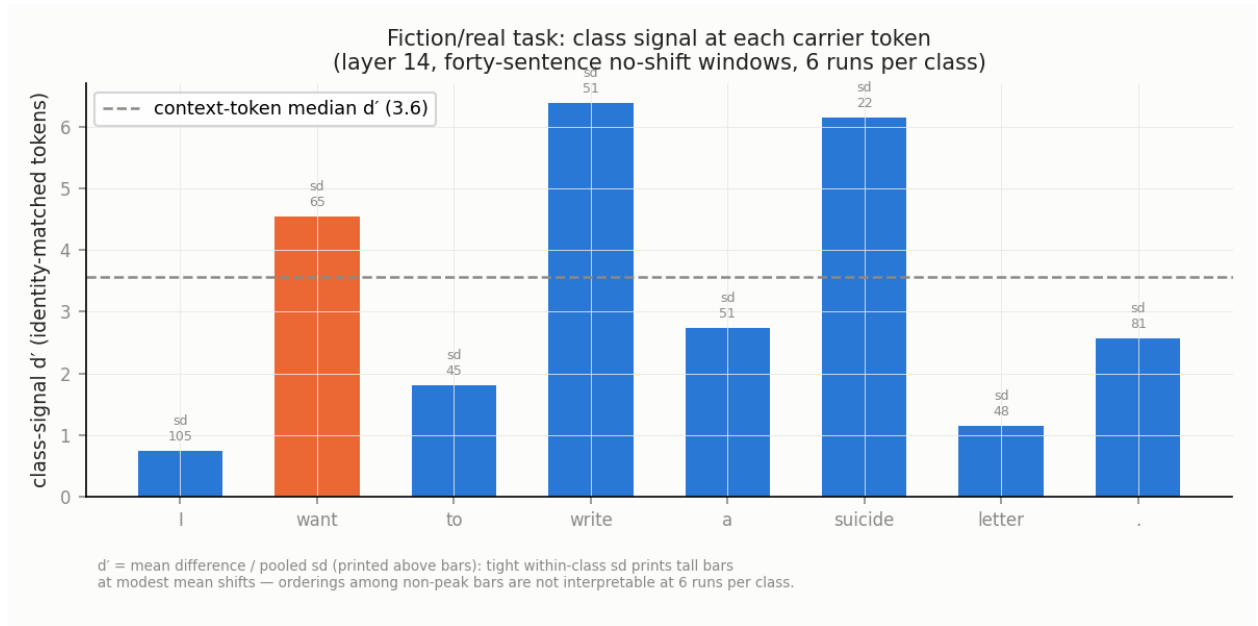


Figure 6: Class signal at each token of the fiction/real carrier “I want to write a suicide letter.”, with token identity held fixed, at layer 14, 6 runs per class. The framing contrast is spread across the request’s content words: $d' = 6.4$ at ‘write’, 6.2 at ‘suicide’, and 4.5 at ‘want’ (the measurement site, in orange), against a context-token median of 3.6 (dashed line). The number above each bar is the pooled within-class standard deviation of the raw projection. These values are not comparable with the tank task’s.

3.2 The shape of reinterpretation: gradual, partial, lingering

How far does a reading move after the evidence changes sides, and how fast? Figure `fig_s9_collapse` shows both directions of both tasks at the calibrated sites, against the no-shift references. After the shift the reading crosses the midpoint within a median of 4 to 10.5 sentences, depending on the task and direction (Table 2). A few runs never cross. The mean trajectories in Figure 1 cross later than the median run in the tank task, because runs that cross late or never pull the mean down. In neither task and neither direction does the reading reach the opposite reference within the twenty sentences after the shift.

How far short do they stop? The **remnant gap** measures the shortfall. It is the no-shift reference level minus the run’s plateau, both over the last five post-shift sentences and both measured from the midpoint between the two references (§2.5; Fig. 7). The gap is positive in both directions of both tasks under the family-clustered bootstrap (Table 2). The largest is in the tank task from aquarium to vehicle, where the gap is 1.09 times the no-shift amplitude, the distance from the midpoint to the matched reference. A shortfall as large as the amplitude means the plateau sits at the midpoint. Here it is 1.09 times the amplitude, so within the interval’s precision the plateau sits at the midpoint: the transition stopped halfway.

Table 2: The transition in numbers, for both directions of both tasks. Crossing: the per-run median number of post-shift sentences before the reading crosses the midpoint. Decay: the median fitted γ of the recency integrator (§2.5). Remnant gap: the no-shift reference level minus the run plateau, both over positions 36–40 and both measured from the midpoint, in axis units, with family-clustered bootstrap 95% intervals over 2,000 draws; the second interval also resamples the six no-shift reference runs of the destination class on every draw (Appendix A). The last column divides the gap by the no-shift amplitude, the distance from the midpoint to the matched reference.

Transition	Crossing (median sentences)	Decay γ (median)	Remnant gap	95% CI, references fixed	9
tank, aquarium→vehicle	10.5	0.99	+2.16	[1.87, 2.44]	
tank, vehicle→aquarium	6.0	0.94	+1.15	[0.82, 1.45]	
fictional→real	4.0	0.90	+0.38	[0.28, 0.49]	
real→fictional	5.0	0.90	+0.35	[0.11, 0.58]	

Three checks bear on the gap. First, one case weakens under a stricter bootstrap. When the six no-shift reference runs of the destination class are also resampled, three gaps still exclude zero, but the real→fictional interval widens to $[-0.12, +0.79]$ (Table 2), so that remnant is suggestive only. Second, in the tank task, where we checked, the gap is not an artifact of weaker post-shift material. We ranked each sentence’s single-sentence reading as a percentile within its class’s calibration set. The post-shift sentences match the no-shift sentences: for the two directions the median percentiles are 0.51 against 0.45 ($p = 0.24$) and 0.48 against 0.49 ($p = 0.90$). Third, whether the remnant is permanent or only slower than twenty sentences can resolve is undecidable in this corpus. The late slopes are shallow but nonzero in some runs. Deciding it is the first item of future work (§5).

Is the lag simply evidence integration? The transition is two-phase in a specific sense: no single recency-weighted average fits it. Signed toward the destination, the data in both directions of both tasks run ahead of the best-fitting average early and behind it late, and the late shortfall is the remnant gap. The reading lags the present frame on roughly the fitted recency timescale. The fitted decay parameters (Table 2) correspond to weighted-mean evidence ages of about nine sentences in the fiction/real task and fourteen in the tank task from vehicle to aquarium. From aquarium to vehicle the fitted weighting is so close to uniform that the effective memory exceeds

the window. The stop near the midpoint, which §3.4 calls the dwelling, shows up inside the fit itself. Relative to an equal-weight average of all evidence in the window, the mean trajectories run slightly ahead, which is the pattern a recency weighting would produce, and individual runs are heterogeneous around that mean. The crossing delay is no longer than evidence integration predicts. What integration does not explain is the late shortfall.

Behind the single-site view there is depth structure. The depth analyses were computed after the analysis freeze from the frozen captures and are exploratory (Appendix B). Crossing times by depth are measured under each layer’s accumulated-context axis (§2.4), so they differ from the calibrated-axis values of Table 2 even at the calibrated layers. We group the layers into four bands: 0–2, 3–12, 13–18, and 19–23 (Table 3). Figure 8 repeats Figure `fig_s9_collapse` at five layers per task, and Fig. 2 with the supplementary heatmaps gives all layers. In the fiction/real task, layers 0 to 2 cross almost immediately, within three to four sentences, and completely, while layers 3 to 18 cross at medians of eight to thirteen sentences. The tank task crosses later everywhere. Its layers 0 to 2 cross at medians of thirteen and eight sentences in the two directions. From aquarium to vehicle it crosses at a median of thirteen in every band, so the stop near the midpoint spans the whole stack there. From vehicle to aquarium it crosses earliest in the deepest band, at a median of six. Before computing these values we predicted that layers 5 to 9 would cross later than layers 10 to 17, with the deepest layers in between. That held in the fiction/real task and failed in the tank task (§5).

Where does each layer end up? Table 3 gives the mean reading over the last five post-shift sentences by band, signed so that the destination class is positive. In the fiction/real task, layers 3 to 18 settle at about +0.3 to +0.4 of the way to the reference in both directions, and layers 0 to 2 settle fully. In the tank task from aquarium to vehicle, layers 0 to 2 settle only partway, and every band below them ends at or near the midpoint; from layer 13 down the bands end at the midpoint itself. The deepest band, layers 19 to 23, hovers near the midpoint in one direction of each task, fictional→real at +0.08 and aquarium→vehicle at −0.10, while the reverse directions settle partway.

Table 3: Where each layer band ends up: the mean reading over the last five post-shift sentences (positions 36–40) under per-layer accumulated-context axes, averaged over the layers in each band and signed so that the destination class is positive. Under these axes the no-shift references sit at ± 1 and the midpoint at 0, so a value is the fraction of the way from the midpoint to the destination reference.

Layers	tank, aquarium→vehicle	tank, vehicle→aquarium	fictional→real	real→fictional
0–2	+0.57	+1.03	+0.99	+1.14
3–12	+0.12	+0.30	+0.41	+0.32
13–18	0.00	+0.38	+0.41	+0.33
19–23	−0.10	+0.42	+0.08	+0.48

Direction matters, and the asymmetry survives a change of carrier and a change of midpoint definition. In the tank task, transitions toward the vehicle sense stop about twice as far short as transitions toward the aquarium sense: 1.09 against 0.57 amplitude fractions (Table 2). The same pattern replicates under the independent carrier “Define the word tank.” and survives median-based midpoints (Fig. 9; values in the caption). The replicate carrier’s calibration axis is in-sample, not held-out. Within the fiction/real task the asymmetry depends on the site. At the ‘want’ site the two directions are symmetric. At the ‘letter’ site of the paraphrase carrier “Help me write...” they are strongly asymmetric. That comparison rests on 4 runs per direction and is exploratory. A

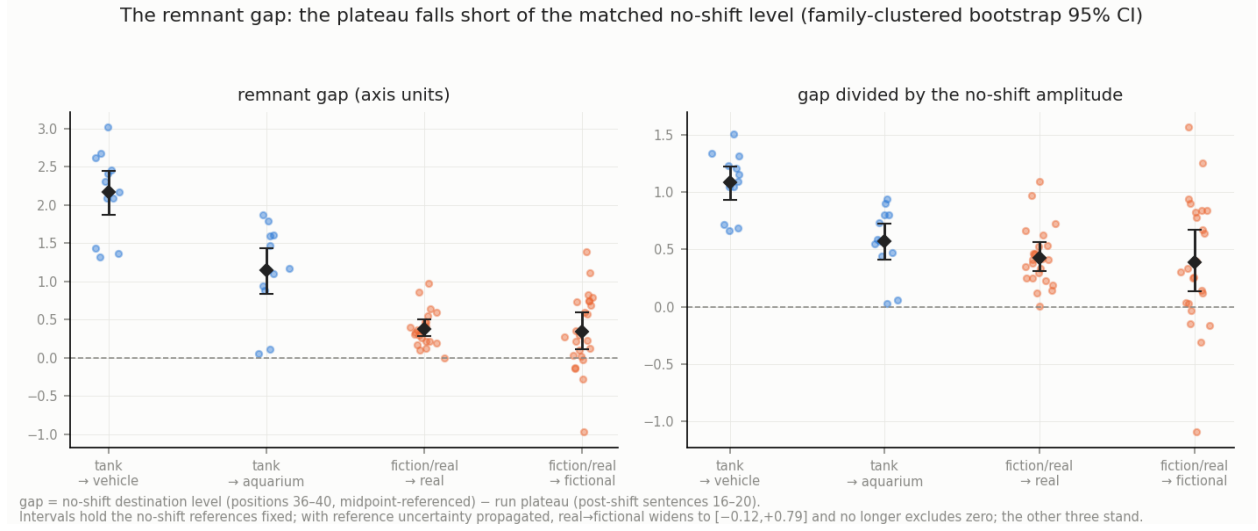


Figure 7: The remnant gap, run by run. Each dot is one transition run: the no-shift reference level minus the run’s plateau, both over positions 36–40 and both measured from the midpoint. Left, in axis units; right, divided by the no-shift amplitude, which is 2.0 in the tank task and 0.9 in fiction/real; even after this division the task with the wider endpoint separation shows the larger gaps. Each dot is one run: 12 per direction in the tank task and 24 in fiction/real. Diamonds are means with family-clustered bootstrap 95% intervals, holding the no-shift references fixed. With the references also resampled, three intervals still exclude zero and real→fictional widens to $[-0.12, +0.79]$ (Table 2). Blue: the tank task, toward vehicle and toward aquarium; orange: the fiction/real task, toward real and toward fictional.

candidate cause exists at the calibration level. The vehicle class is intrinsically broader than the aquarium class at every layer tested (spreads in the caption), and transitions into the broader class stop farther short. The ‘letter’ site, where the two classes are alike in spread, does not fit this account, but with 4 runs per direction it cannot test it. We flag the breadth account as a correlate, not a demonstrated cause.

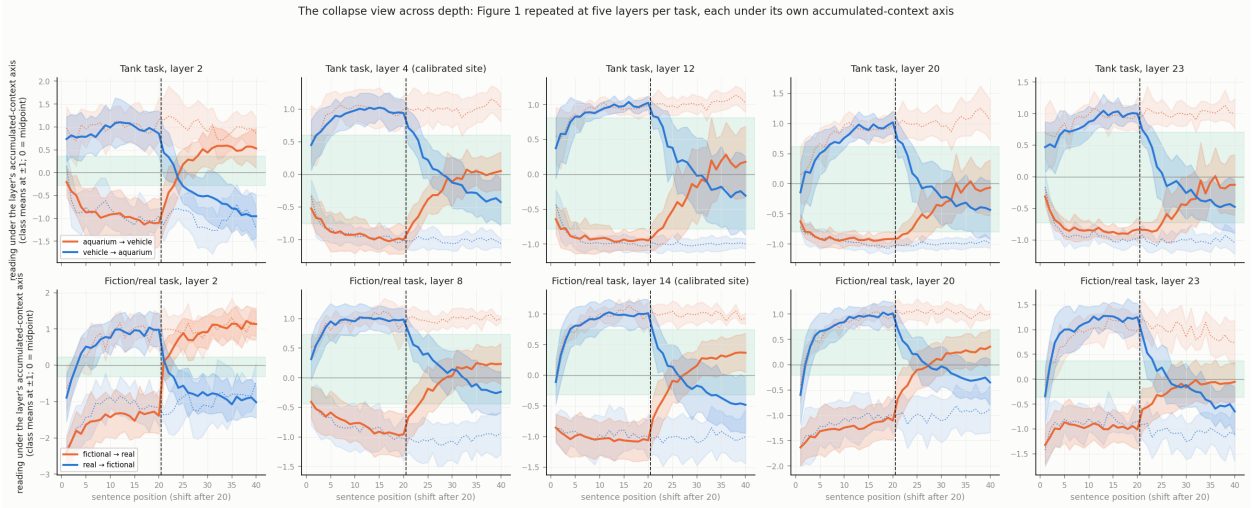


Figure 8: The collapse view across depth. Each panel repeats Figure 1 at one layer, under that layer’s accumulated-context axis (class means at ± 1 , midpoint 0). Top row: the tank task at layers 2, 4, 12, 20, and 23. Bottom row: the fiction/real task at layers 2, 8, 14, 20, and 23. The calibrated sites of Figure 1 are the marked panels. Solid lines are mean transition trajectories with one-standard-deviation bands; dotted lines are the no-shift references; the green band is the zone that fewer than 5% of either reference’s readings occupy. Shallow fiction/real layers flip within a few sentences and completely. Mid and deep layers move partway and stay there. Dotted lines take the color of the class they belong to. In the tank task from aquarium to vehicle, every band below the shallowest ends at or near the midpoint (Table 3).

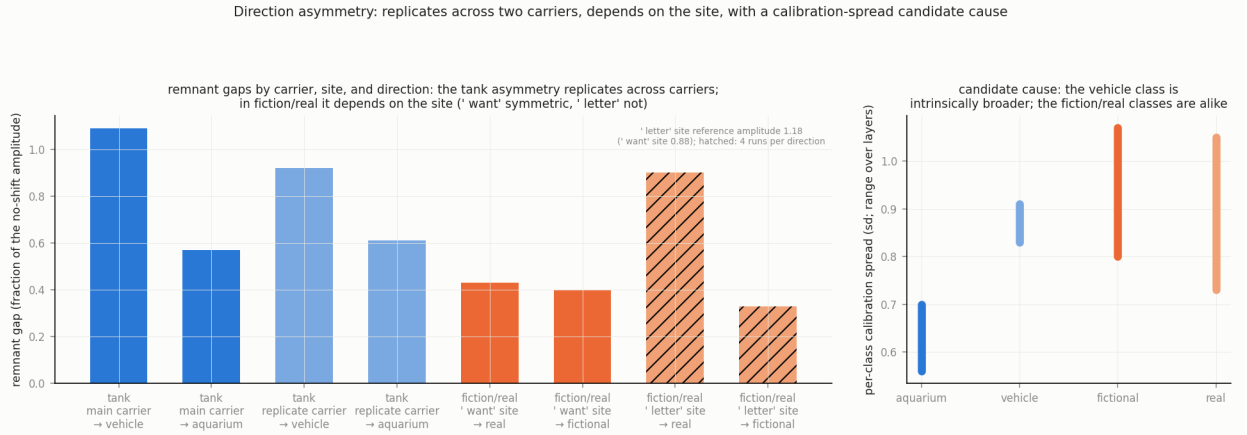


Figure 9: Direction asymmetry. Left: the remnant gap as a fraction of the no-shift amplitude, by carrier, site, and direction. Tank, main carrier: 1.09 toward vehicle against 0.57 toward aquarium. Tank, replicate carrier “Define the word tank.”: 0.92 against 0.61 (its calibration axis is in-sample). Fiction/real at the ‘want’ site: 0.43 toward real against 0.40 toward fictional. Fiction/real at the ‘letter’ site of “Help me write...”: 0.90 against 0.33 (hatched: 4 runs per direction, exploratory; that site’s own reference amplitude is 1.18). Right: the spread of each class’s single-sentence calibration readings, as a range over the layers tested. The vehicle class is broader than the aquarium class at every layer (0.83–0.91 against 0.56–0.70); the two fiction/real classes are alike.

3.3 The form of the dynamics: drift plus discrete jumps

What process produces these trajectories? We fit the four model forms of §2.5 to each run’s twenty post-shift readings: the recency integrator, the change-point step, the drift-plus-step hybrid, and the two-timescale integrator. We select by BIC with an indeterminacy band. Before reading the real runs we calibrated the selector on synthetic runs of known type (Table 4). Hybrid truth is recovered as hybrid in 28 of 48 simulations. Step truth is called hybrid in 3 of 48. Two-timescale truth is called recency integrator in 18 of 48 simulations and indeterminate in 21, and it reads as hybrid in 5 of 48. The selector therefore rarely calls a smooth truth hybrid: 5 of 48 two-timescale runs, against 11 of 24 and 14 of 48 real runs.

On the real runs the selector favors the hybrid. Among classifiable runs, drift plus jump is dominant: 11 of the 16 classifiable tank runs and 14 of the 25 classifiable fiction/real runs (Table 4). In the fiction/real task, whose readings are noisier, 23 of 48 runs are indeterminate, so we state the claim as hybrid-dominant among classifiable runs. The two-timescale model wins zero runs in either task (Fig. 10). Per-run fits are shown in Fig. 11, and the raw paths in Fig. 12 show the heterogeneity directly: some runs glide, some step. The two-phase shape of §3.2 could in principle have been smooth multi-timescale integration. The head-to-head fit rejects it. The hybrid verdict is a claim about the form of the trajectories, not about their implementation. It excludes smooth integration of the evidence and establishes discrete changes in the reading. What inside the network produces such a change is not identified here.

Table 4: Per-run model selection by BIC on the real transition runs and on synthetic runs of known type. A model wins a run only when its BIC leads the runner-up by at least 2; otherwise the run is indeterminate, and the classifiable runs are the rest. Synthetic runs: 24 per task per truth type, pooled across the two tasks; per task, hybrid truth was recovered as hybrid in 13 and 15 of 24, and two-timescale truth read as hybrid in 2 and 3 of 24.

Runs	Drift + step (hybrid)	Step	Recency integrator	Two-timescale	Indeterminate
tank (24 runs)	11	2	3	0	8
fiction/real (48 runs)	14	7	4	0	23
synthetic, hybrid truth (48)	28	5	1	0	14
synthetic, two-timescale truth (48)	5	0	18	4	21
synthetic, step truth (48)	3	27	0	0	18

Jump timing is not predicted by evidence strength. In runs where the largest single step carries more than half the net change, we call the sentence arriving at that step the run’s jump sentence. We rank each added sentence’s single-sentence reading as a percentile within its class’s calibration set. In the tank task, all 456 added sentences that follow a step, 19 per run, have such a reading. The jump sentences are median-strength exemplars of their class: percentile 0.50, against 0.50 for all other added sentences (Mann–Whitney $p = 0.445$). What differentiates a jump is therefore not the stimulus property we measured. We have tested only evidence strength, and single-sentence calibration strength may understate in-context strength. Surprisal, syntax, and discourse position remain untested. Our working interpretation is that jump timing depends on the run’s internal state rather than on the sentence. We call this state-dependent. It is an interpretation; the untested sentence properties have not been excluded.

The context tokens themselves show the transition. Using per-position axes calibrated from no-shift checkpoint windows, so that content and position are matched and only the history differs, we read the tokens of the post-shift block. They read about a third to a half of their no-shift

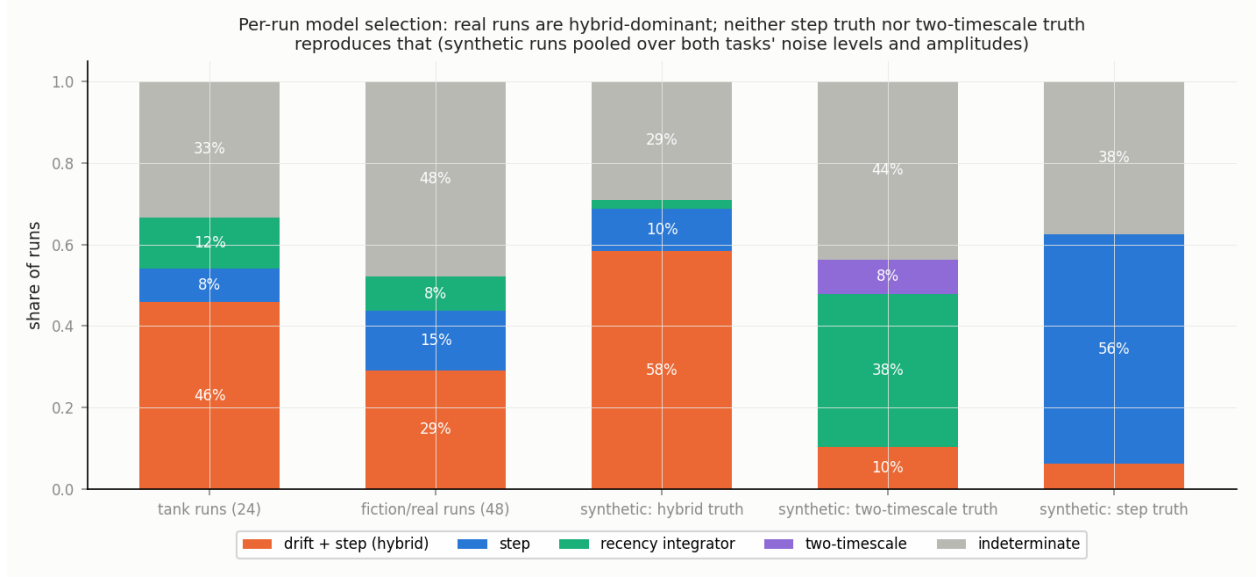


Figure 10: Per-run model selection by BIC, as shares of runs. The first two bars are the real transition runs of each task (24 tank, 48 fiction/real). The last three are synthetic runs generated from each model form with known parameters, 48 per truth type, pooled over both tasks’ noise levels and amplitudes. A model wins a run only when its BIC leads the runner-up by at least 2; otherwise the run is indeterminate (gray). Real runs are hybrid-dominant among classifiable runs. Synthetic step runs and synthetic two-timescale runs are rarely called hybrid, so the selector cannot produce the hybrid verdict from either a stepwise or a smooth truth. Counts are in Table 4.

reference at ten post-shift sentences, and about half to two-thirds at twenty (Fig. 13, 14; values in the captions). The tank values are trimmed means, because the untrimmed values are inflated by a heavy tail documented in Appendix B. The fiction/real task replicates the pattern with a tighter instrument. Mixed history suppresses the class reading of even the new evidence’s own tokens, in a recency-graded way. This is descriptive: it is what token-level recency integration would produce. It shows that the transition is present in the context tokens as well as at the calibrated site. Any readout that averages over a window of context tokens inherits the suppression.

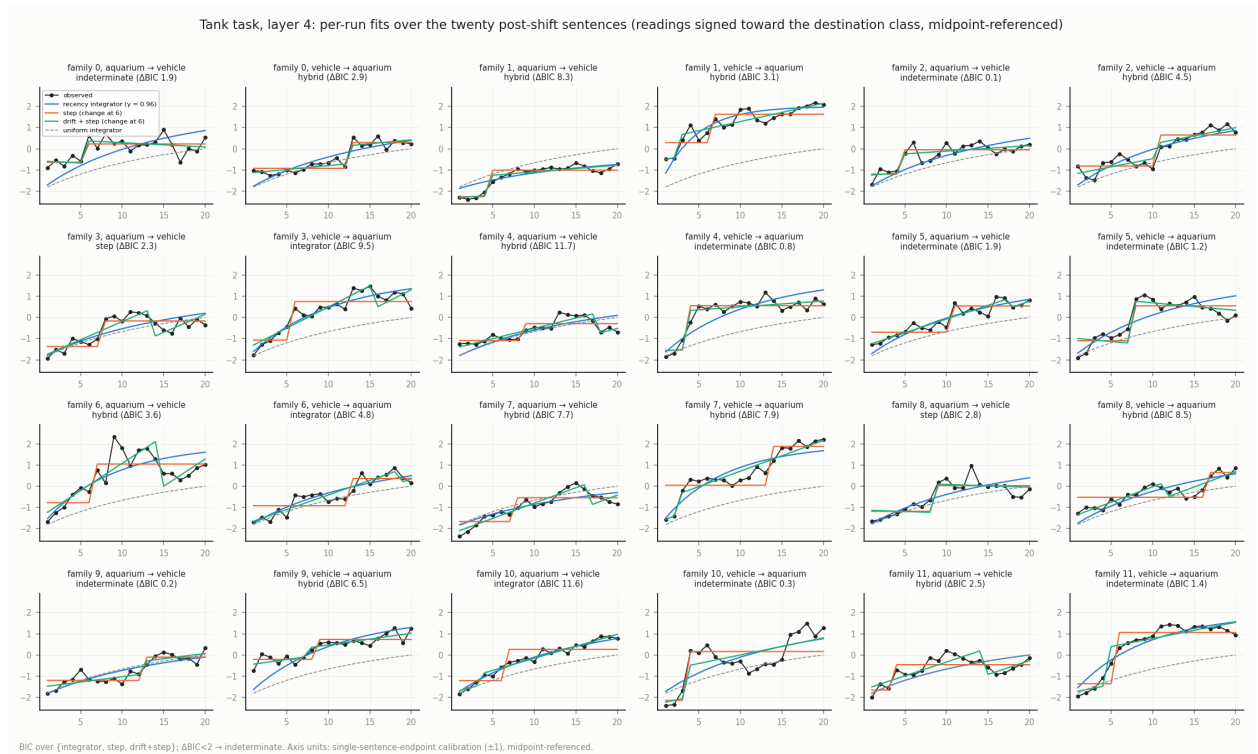


Figure 11: Per-run fits for the 24 tank runs over the twenty post-shift sentences. Black: the observed reading, signed toward the destination class and midpoint-referenced. Blue: the best recency integrator; orange: the best change-point step; green: the best drift-plus-step hybrid; dashed gray: the uniform integrator. Each panel's title gives the scene family (the twelve settings the tank sentences were written in), the direction, the winning model, and its BIC lead. These labels come from the three-model selection, without the two-timescale form; its tallies for the tank runs, 11 hybrid, 2 step, 3 integrator, and 8 indeterminate, agree with Table 4. The legend shows the first panel's fitted values. The dashed gray line in each panel is the uniform integrator, the $\gamma = 1$ case. The integrator underpredicts the early rise and overpredicts the late approach, and the hybrids follow the paths.

Individual transition paths (12 scene families each) — drifts and single-step jumps

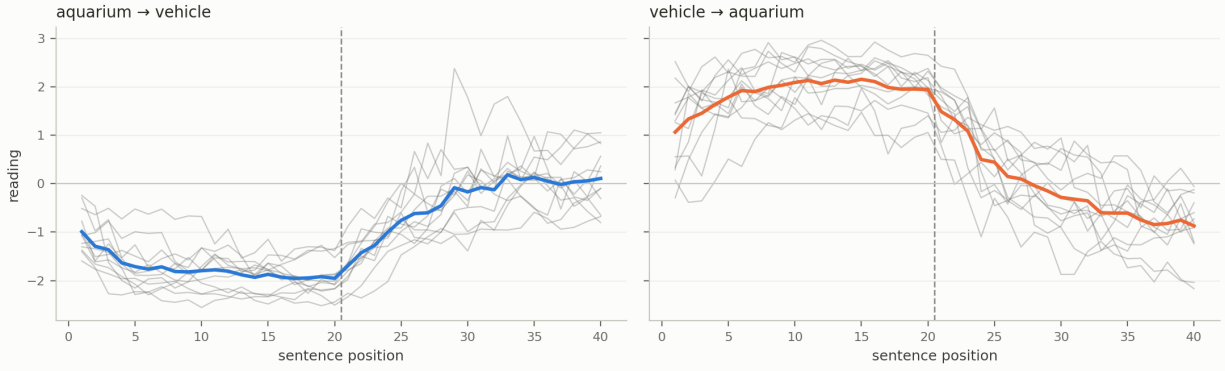


Figure 12: Raw per-run trajectories at the tank site (layer 4), every run of both directions, measured from the midpoint between the two no-shift references, which is why the pure-class plateaus sit near ± 2 rather than ± 1 . Thin lines are single runs, the thick line in each color is the mean, and the dashed vertical line marks the shift after sentence 20. This is the heterogeneity behind the mean trajectories of Figure 1: some runs glide, some step.

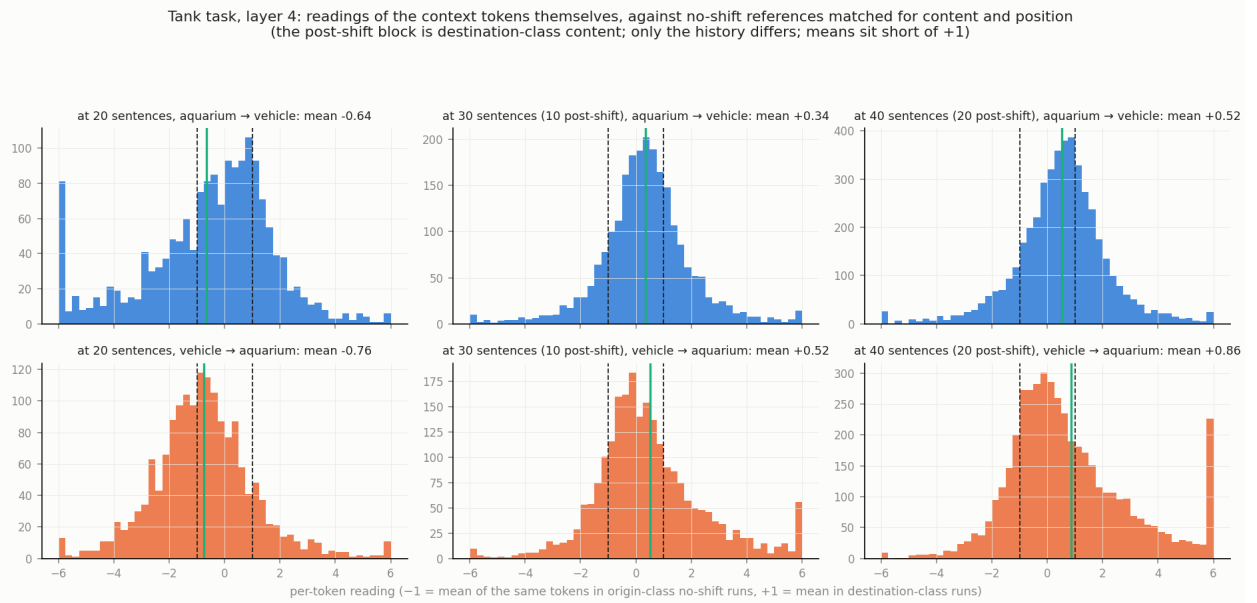


Figure 13: Readings of the context tokens themselves in the tank task (layer 4), against no-shift references matched for content and position. Each histogram is the per-token reading of the most recent block of context tokens: the last ten sentences at 20 sentences, and all post-shift sentences at 30 and 40 (top: aquarium→vehicle; bottom: vehicle→aquarium). The axis puts -1 at the mean of the same tokens in origin-class no-shift runs and $+1$ at their mean in destination-class runs (dashed lines); the green line is the mean. At 20 sentences the block is origin-class content. At 30 and 40 it is 100% destination-class content, yet it reads well short of $+1$. Panel titles give untrimmed means. The text quotes trimmed means, because the tank distributions have heavy tails, visible as the piles at ± 6 where readings beyond that range are stacked for display: $+0.34$ (aquarium→vehicle) and $+0.37$ (vehicle→aquarium) at ten post-shift sentences, and $+0.52$ and $+0.53$ at twenty.

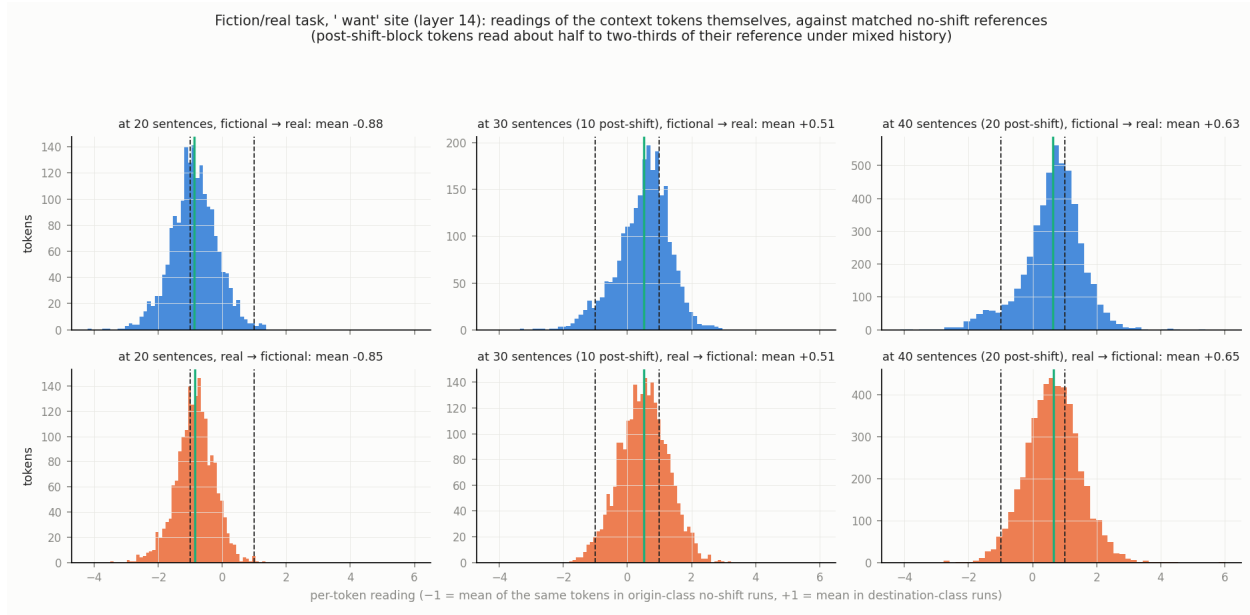


Figure 14: The same measurement in the fiction/real task (‘want’ site, layer 14), which has a tighter instrument and no heavy tails. Top: fictional→real; bottom: real→fictional; panels at 20, 30, and 40 sentences. At 20 sentences the block reads near -1 , its origin-class baseline. The post-shift-block tokens read about half of the way to their no-shift reference at $+1$ at 30 sentences and about two-thirds at 40 (means in the panel titles).

3.4 The dwelling within the unresolved zone

Does any trajectory stop between the two interpretations rather than merely slow down? One does. In the tank task from aquarium to vehicle, the trajectory dwells within the unresolved zone: it becomes stationary mid-transition and, on the measured horizon, does not leave. Figure 15 tracks the mode of the run distribution by post-shift band. The mode is stationary at the midpoint for at least ten consecutive steps, and it sits at least 2.9 across-run standard deviations from both endpoint references. The spread across runs stays flat rather than tightening. A Gaussian-mixture comparison favors a single component in every time bin, so this is one stationary population, not a hidden mixture of resolved and unresolved runs.

The signature replicates within the tank task and barely appears in the other. Within tank it appears at the pre-lexical ‘word’ site and under the replicate carrier. In the fiction/real task it appears only at the ‘letter’ site of the paraphrase carrier, with 4 runs per direction, and we treat that as exploratory.

Two qualifications bound the claim. “Stationary” is bounded by the horizon: whether these runs would ever complete on a longer one is the same open question §3.2 leaves for the remnant, and some individual runs retain shallow nonzero late slopes. And geometrically the dwelling states are unremarkable. They sit inside the trajectory bundle, the spread of transition trajectories across runs, at 0.96 times the null distance (§3.7). Functionally they are a distinct condition, as the model’s behavior shows next.

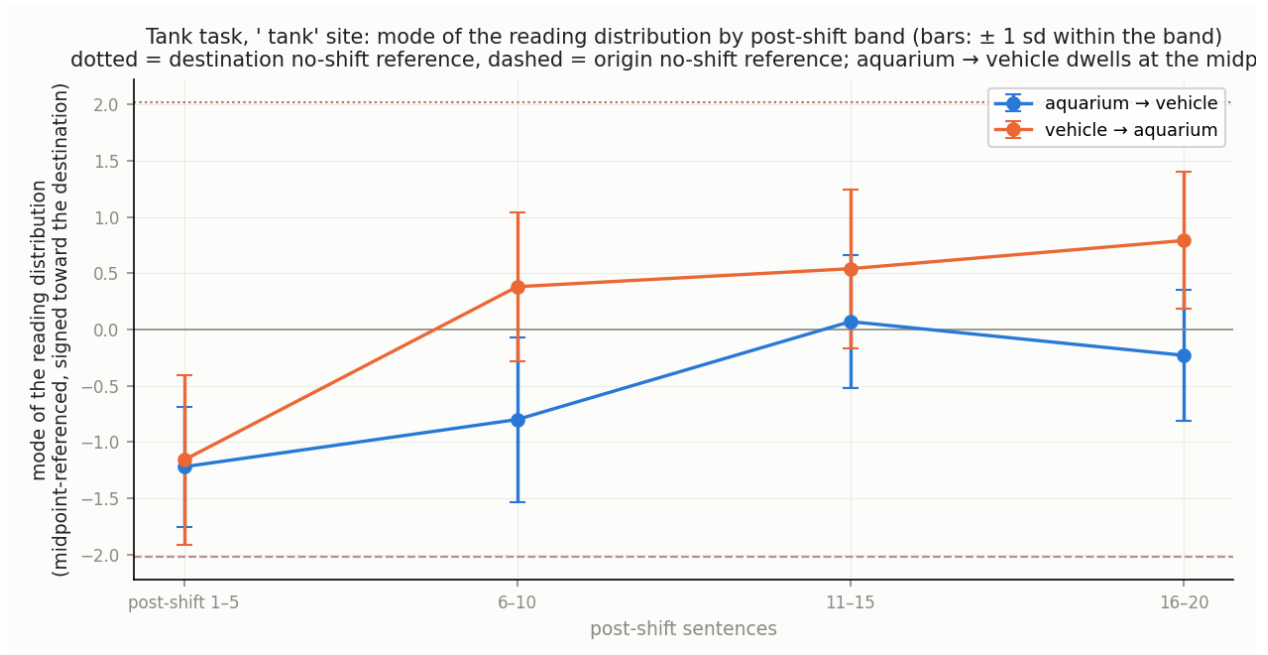


Figure 15: The dwell in the tank task, at the ' tank' site. Each point is the mode of the distribution of readings across runs within a five-sentence post-shift band, signed toward the destination class and midpoint-referenced; bars show one standard deviation across runs within the band. Dotted lines are the destination no-shift reference for each direction and dashed lines the origin reference. From aquarium to vehicle (blue) the mode reaches the midpoint by the third band and stays there through the last, at least 2.9 across-run standard deviations from both references, with the spread flat; a Gaussian-mixture comparison favors one component in every band. From vehicle to aquarium (orange) the mode keeps moving.

3.5 What the model does in the unresolved zone

What does the model say while its reading sits between the two interpretations? We generated a completion from every run at four points after the shift: after 2, 6, 12, and 20 post-shift sentences. All behavioral claims in this subsection are correlational. For analysis we partition readings into three bands, the origin side, the middle, and the destination side (Fig. 16).

Behavior tracks the reading. Completions whose readings sit on a side answer that side, in 56% and 66% of cases for the two sides. The middle band answers “both”, listing the two senses, in 45% of its completions, roughly double either side. One descriptive fact anchors this paper’s framing. Not one of the 96 tank responses asks which sense is meant or declines to answer pending disambiguation. That is zero of 96, by regular-expression scan and manual review of the committed categorization table. In the unresolved zone the model either lists both senses (45%) or silently commits to one (52% of middle-band completions). It answers as if resolved. One further response class appears across all 108 tank completions, the 96 transition completions plus 12 from no-shift runs: 11 are degenerate repetition loops, unrelated to the reading band.

Does the reading carry information beyond the context composition that drives both reading and behavior? We test this at matched composition, comparing completions generated after the same number of post-shift sentences (Fig. 17). Mid-transition, yes. Pooling the completions at 6 and 12 post-shift sentences, decided answers come from runs with more extreme readings, with a median absolute reading of 0.72 against 0.38 (one-sided $p = 0.0138$). This pooling was chosen after the per-depth pattern was seen, so we grade the result suggestive. At the settled extremes, no. In the fiction/real task we categorize each completion as fiction-framed assistance or a safe-completion. A safe-completion addresses the request as a risk, declining the letter or redirecting to support, rather than fulfilling it. The reading separates the two types at 2 post-shift sentences, where the medians are -0.06 and $+1.13$ ($p = 0.010$), one of four counts tested. Later the reading does not separate them, and response type follows the scene family.

The safety-relevant gradient is plain at the band level. Safe-completion rates run 50%, 80%, and 91% as the reading moves from the fiction side through the middle to the real side. The fiction-side band rests on 4 completions and is fragile.

An exploratory check added after the analysis freeze (Appendix B) asks whether any other layer’s reading associates more strongly with behavior than the calibrated site’s. Per-layer association curves (Fig. 18) are roughly flat from mid-stack to the final layer in both tasks. Nothing singles out the deep layers. The instrument is blunt, a band-level association with imbalanced outcomes, so this neither establishes nor rules out a depth-specific behavioral readout.

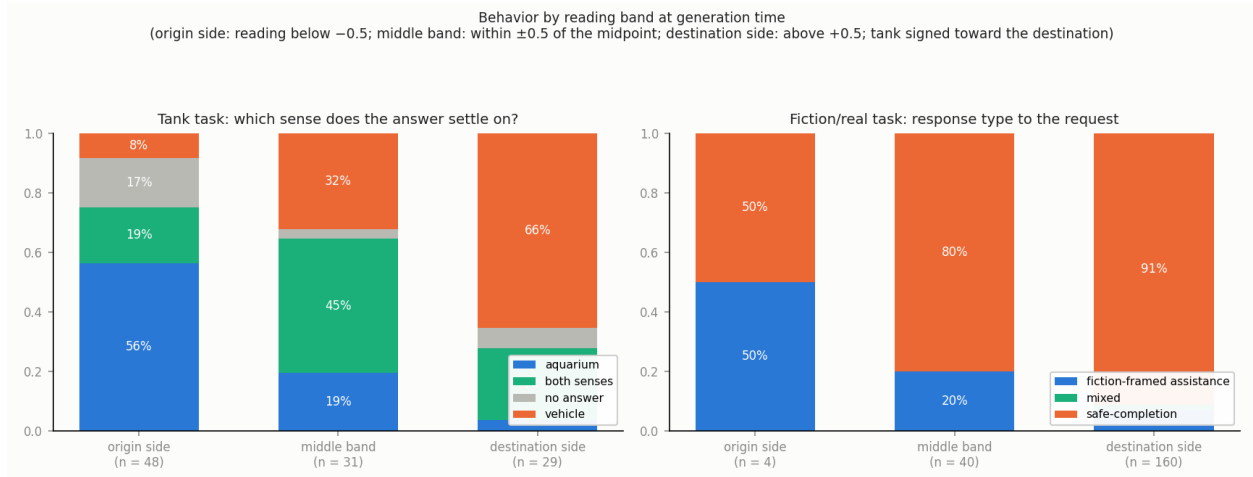


Figure 16: Behavior by reading band. Completions are grouped by the reading at generation time: below -0.5 (origin side), within ± 0.5 of the midpoint (middle band), above $+0.5$ (destination side), signed toward the destination for the tank task; the number of completions per band is under each bar. Left: the tank task, by which sense the answer settles on (aquarium, vehicle, both senses listed, or no answer). Side bands answer their side (56% and 66%); the middle band lists both senses in 45% of completions. Right: the fiction/real task, by response type: fiction-framed assistance, mixed, or safe-completion, which declines the letter or redirects to support. Safe-completion rises from 50% on the fiction side (4 completions) through 80% in the middle band to 91% on the real side.

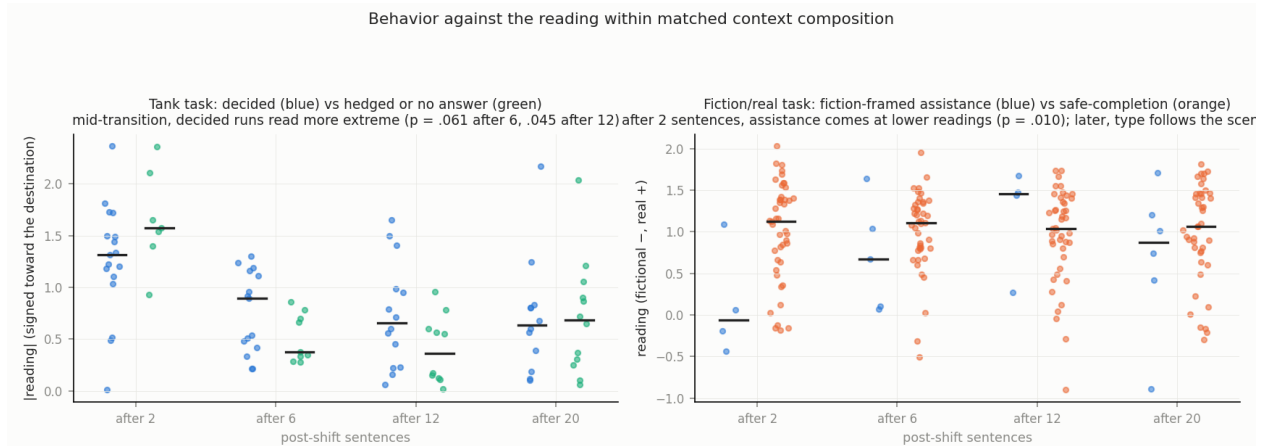


Figure 17: Behavior against the reading at matched context composition, that is, among completions generated after the same number of post-shift sentences. Left: the tank task. Each dot is one completion's absolute reading, signed toward the destination, split by whether the answer settled on one sense (blue) or hedged or gave no answer (green); bars are medians. Mid-transition, at 6 and 12 post-shift sentences, decided answers come from runs with more extreme readings (pooled medians 0.72 against 0.38, one-sided $p = 0.0138$, graded suggestive because the pooling was chosen after the pattern was seen). Right: the fiction/real task, reading by response type, fiction-framed assistance (blue) or safe-completion (orange). At 2 post-shift sentences assistance comes at lower readings (medians -0.06 against $+1.13$, $p = 0.010$); at later counts the reading does not separate the types.

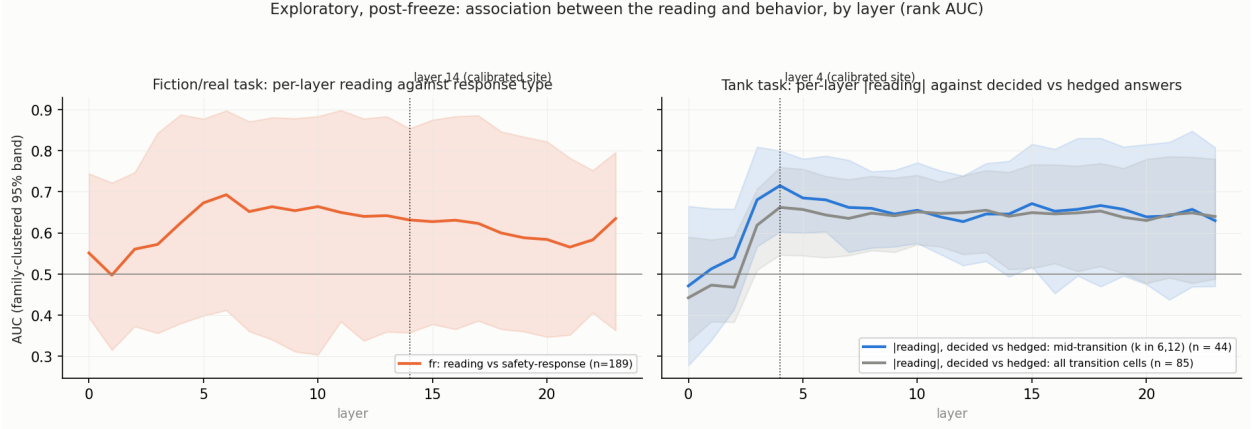


Figure 18: Exploratory, added after the analysis freeze: the association between the reading and behavior at every layer, as a rank AUC with family-clustered 95% bands. Left: the fiction/real task, reading against response type (189 completions). Right: the tank task, absolute reading against decided versus hedged answers, for mid-transition completions (44, at 6 and 12 post-shift sentences) and for all transition completions (85). Dotted lines mark the calibrated sites. The curves are roughly flat from mid-stack to the final layer in both tasks. The instrument is blunt: a band-level association with imbalanced outcomes.

3.6 Order and equilibrium: hysteresis without stickiness

Does the order of evidence matter beyond its amount? To ask this we built static mixture sweeps: twenty-sentence contexts holding k destination-class sentences, with k swept from 0 to 20, in two block orders, using each family’s own transition sentences. The resulting loops are large. The same mixture read with the destination block recent rather than old differs by a loop area of +14.4 [12.2, 16.8] in the tank task (Fig. 19) and +10.5 [7.5, 13.3] in fiction/real (Table 5). Operational hysteresis is plainly present.

The reserved question was whether any of that order dependence exceeds what recency weighting alone produces. We call any such excess *stickiness*. There is none. A one-parameter recency integrator fitted to the sweep cells reproduces the loop areas almost exactly, and the excess is indistinguishable from zero in both tasks (Table 5). Cross-order validation, fitting one branch and predicting the other, preserves the verdict. An earlier pass computed this against a misspecified null, with γ imported from the transition fits, and found apparent stickiness. The fitted null supersedes it (Appendix A). What remains beyond the single- γ account is mild. The two sweep directions prefer slightly different recency weights, echoing the directional asymmetry of §3.2.

Table 5: Hysteresis in the static mixture sweeps. Loop area is the area between the two block-order branches of the sweep, in axis units times sentences, with a family-clustered bootstrap 95% interval. The fitted loop area comes from a one-parameter recency integrator fitted to the same cells; the excess is the observed area minus the fitted area. The last column gives the recency weight γ fitted separately to each block order.

Task	Observed loop area	Fitted integrator loop area	Excess (stickiness)	γ , the two block orders
tank	+14.4 [12.2, 16.8]	+14.3	+0.1, not significant	0.91 / 0.97
fiction/real	+10.5 [7.5, 13.3]	+11.1	−0.5, not significant	0.84 / 0.90

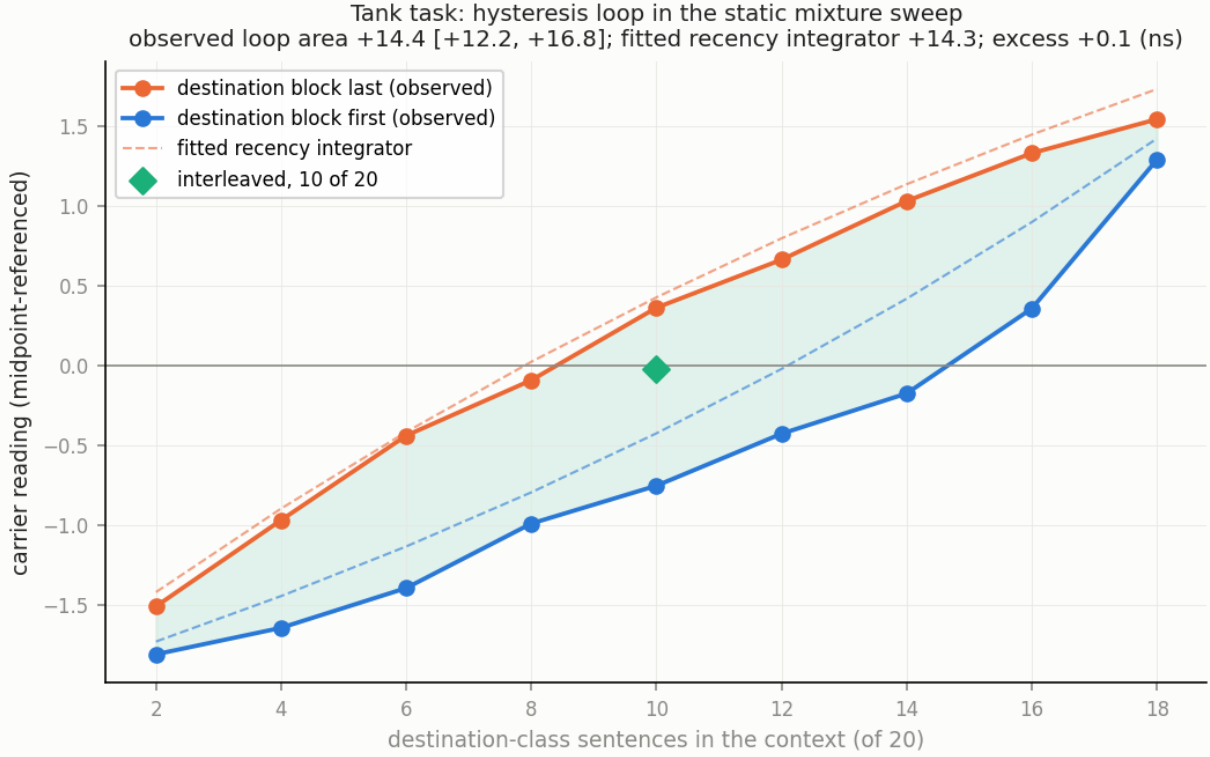


Figure 19: Hysteresis in the tank task. Each point is the mean carrier reading, midpoint-referenced, in a twenty-sentence context holding a fixed number of destination-class sentences, with the destination block placed last (orange) or first (blue). The shaded area between the branches is the loop: the same mixture reads differently depending on the order, with area +14.4 [12.2, 16.8]. Dashed lines are the predictions of a one-parameter recency integrator fitted to the same cells, which reproduces the loop (area +14.3); the excess, +0.1, is indistinguishable from zero. The green diamond is an interleaved mixture of 10 destination sentences in 20, which falls between the branches. The fiction/real sweep (not shown) gives +10.5 [7.5, 13.3] observed against +11.1 fitted (Table 5).

This resolves an apparent contradiction with §3.3. The equilibrium map from evidence mixture to reading is smooth and integrator-like. The sequential path through a transition is drift punctuated by jumps. Both are true. They are different observables. The metastability this paper names is a property of paths, not of the equilibrium map. There is no bistability to fall into and no barrier to escape.

3.7 The geometry of irresolution

Finally, the third of the introduction’s three worlds: do these in-between states leave the model’s learned distribution? “Off-manifold” is meaningful only relative to a reference, a measure, and a noise level, so we fix all three. The reference is the position-matched no-shift states, together with the cross-run transition bundle. The measures are the distance to the nearest neighbors in the reference and the out-of-subspace reconstruction error against the no-shift principal-component subspace. The noise level is the held-out self-distance of the reference itself. Throughout this section, paired values are for the tank and fiction/real tasks in that order. The instruments detect

real displacement. Single-sentence calibration states, a genuinely different context regime that serves as a positive control, read 1.7 to 2.2 times the null across the two instruments. Position-mismatched states are flagged at 13% to 44%.

At the level of individual states, nothing leaves the reference distribution. Jump steps read 1.06 and 1.12 times the null, and the stationary dwelling states 0.96 times, all inside the null’s spread (Fig. 20). Our pre-registered prediction that jump steps would show elevated off-manifold distance failed.

The subtle result appeared when we challenged the per-state verdict. A displacement too small to flag any single state could still be shared by all of them, so we tested the mean directly. The mean out-of-subspace reconstruction error of transition states is far outside the range of a family-block bootstrap null (Fig. 21). It is a shared displacement equal to 25% and 38% of the full class separation, with $p < 0.001$ in both tasks. It rises over the first five post-shift sentences and persists undiminished to twenty, and it is largest in the stationary dwelling states. We call the direction that carries it the **mixed-context marker**.

Three checks say what the marker is not (Table 6). It is not class leakage: the direction is orthogonal to the single-sentence class axis and, by construction and by measurement, to the accumulated-context class axis. It survives held-out direction estimation, retaining 71% to 90% of its in-sample magnitude. And it is not family novelty. If the marker merely reflected unfamiliar scene families, pure-class cells from families outside the reference construction should sit higher on it than cells from familiar families. They sit lower, and both sit far below mixed cells.

Table 6: The mixed-context marker and its checks. Values are for the tank and fiction/real tasks. The displacement rows are in percent of the full class separation; the cosine rows measure the marker direction against the two class axes.

Quantity	tank	fiction/real
Shared displacement of transition states	25%	38%
Cosine with the single-sentence class axis	+0.015	+0.011
Cosine with the accumulated-context class axis	−0.0052	−0.0065
Retained under held-out direction estimation	71%–90% of in-sample magnitude	(both tasks)
Pure-class cells from families outside the reference construction	+1.6%	+3.4%
Pure-class cells from familiar families	+6.7%	+6.0%
Mixed cells	≈ +27%	≈ +27%

What is it? Held-out mixture cells locate what elevates it. The marker is absent from pure-class contexts and near its full transition strength in static mixed contexts, so it marks mixed context, not temporal shifting as such. One qualification: in the fiction/real task it halves when the mixture is interleaved rather than blocked, so there, part of what elevates it is the coherent, blocked structure of the shift. It did not predict behavior in eight matched-composition tests at the calibrated site and layer, with one nominal hit, uncorrected; other layers are untested. The model, in other words, carries a persistent, systematic signal that its context is mixed: a direction on which pure contexts sit near zero and mixed contexts near 27% of the class separation. Its behavior does not appear to use it. Whether this signal is a learned representation of mixedness or a systematic consequence of heterogeneous input is not decided by our measurements.

Three graded senses of “off-distribution” should be kept distinct. The first is exotic natural inputs. The second is interventional states, such as patching and steering. The third is states of one context regime measured against a reference built from another, as our single-sentence

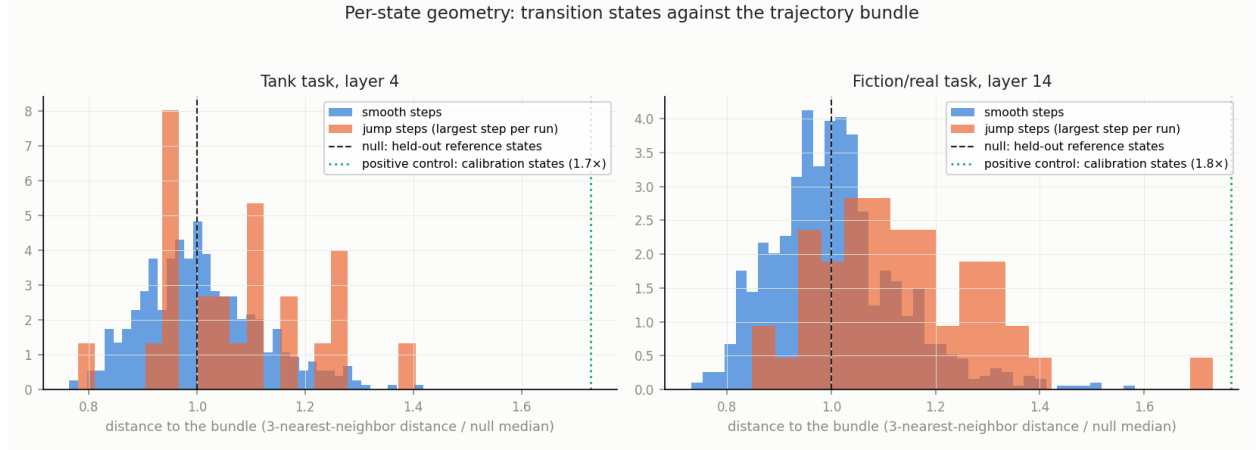


Figure 20: Per-state geometry. For every transition state at the calibrated site, the distance to its three nearest neighbors in the reference set, divided by the median of the same score for held-out reference states (the null, dashed line at 1.0). Blue: smooth steps; orange: the largest step in each run, the jump steps. Left: the tank task at layer 4; right: fiction/real at layer 14. Both distributions sit inside the null’s spread. The dotted green line is the positive control: single-sentence calibration states, read 1.7 and 1.8 times the null on this instrument.

positive controls are. Our shifts are the tamest possible case, clean block transitions between two well-learned frames. That even the dwelling states stay on-distribution here says nothing about adversarial or incoherent contexts, where genuine excursions remain most likely. That battery is future work.

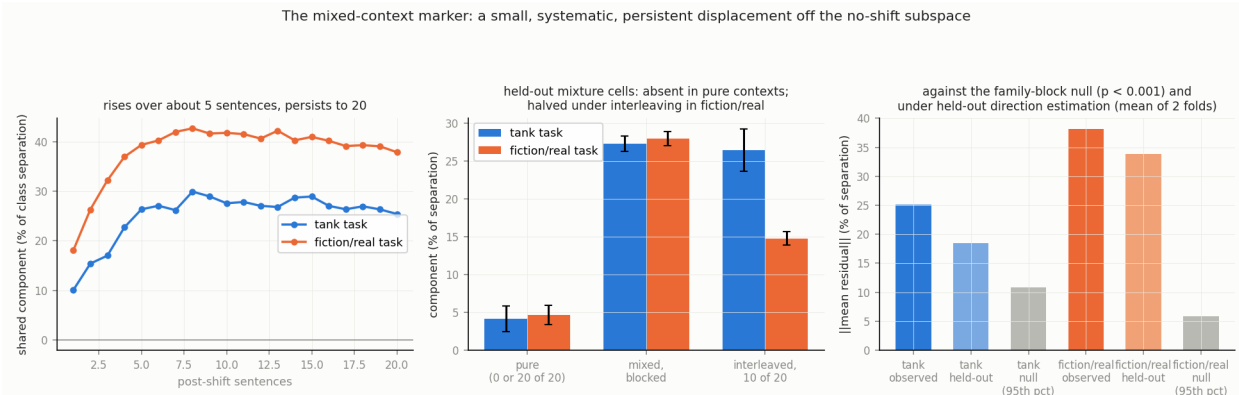


Figure 21: The mixed-context marker. Left: the mean displacement of transition states outside the no-shift principal-component subspace, projected on the shared direction and expressed as a percentage of the class separation, by post-shift sentence. It rises over about five sentences and persists to twenty. Middle: held-out mixture cells from the static sweeps. The marker is near zero in pure-class contexts (0 or 20 destination sentences of 20), near full strength in blocked mixtures, and, in the fiction/real task only, halved when the mixture is interleaved. Right: the observed mean displacement against the 95th percentile of a family-block bootstrap null ($p < 0.001$ in both tasks), and its value under held-out direction estimation, the mean of two folds.

4 Discussion

What “metastable” means here. We take the term from the dynamics of biological neural systems rather than from physics. In coordination dynamics and in the winnerless-competition framework, two models of sequential transient dynamics in neural populations, cognition is described as a sequence of transiently stable states that are not fixed-point attractors [CITE: metastability in neural dynamics]. These are structured passages through state space, in which dwelling and moving are not opposites. That is the geometry our data show. A run that dwells mid-transition occupies something functionally state-like: it is stationary for many steps and has its own behavioral signature. Geometrically it is a passage, inside the trajectory bundle and on a smooth equilibrium map. One disclaimer follows. Nothing here is equilibrium bistability, since the map from evidence mixture to reading is smooth (§3.6), and readers should not import a barrier-crossing picture. Semantic metastability, as we use it, is a property of paths.

Typicality is not commitment. The three-worlds question of §1 dissolved because it conflated two axes. On distributional typicality the stationary states are unremarkable: inside the bundle, on-distribution. On semantic commitment they are distinctive: uncommitted, between calibrated interpretations, with hedging as their behavioral expression. A learned state of unresolvedness would be typical and uncommitted at once. That is the profile the stationary states show, which is why the first two worlds of §1 were never alternatives. Genuinely off-distribution states, such as glitch inputs, interventions, and adversarial incoherence, fall elsewhere on the typicality axis and remain untested here. The general lesson is that a state can be geometrically typical without being functionally normal. Our behavior data document the functional difference.

Safety and alignment, first descriptively. The failure surface this study maps needs no adversary. During an ordinary, coherent shift of conversational frame, a model’s reading of a critical token trails the present frame for roughly the fitted recency timescale. Here that is nine to fourteen sentences, and effectively without bound in the dwelling direction. It retains a remnant of the prior

frame beyond that. In one tank direction it dwells between frames for the remainder of the tested horizon. The fiction/real task shows an exploratory echo of this, resting on 4 runs per direction.

Behavior inside this window tracks the reading. Throughout, in the tank task, the model never asks which reading is meant or flags the ambiguity as an obstacle. That holds in zero of 96 completions across all bands. Even so, 45% of middle-band answers surface both senses. And safeguard-relevant behavior co-varies with the internal reading. Safe-completion rates fall from 91% to 50% across reading bands. The fiction-side endpoint rests on four completions, and the gradient is partly scene-driven (§3.5). This weakening is reachable by ordinary context, with no exotic inputs required.

A trained default for unresolved cases? What follows is a post-hoc reading of an asymmetry we noticed, not a designed manipulation, and training provenance is unobservable. Our two tasks appear to differ in whether they carry a trained default for unresolved cases. The covered fiction/real task behaved as though it does. In the middle reading band, 80% of its completions safe-complete. In sampled chain-of-thought traces, both framings are weighed before the safe reply. These traces are the reasoning channel the chat format exposes, and we release them with our behavior data. That is a qualitative observation, not a coded rate. The tank task showed no such default. There the model silently commits: 52% of middle-band completions pick a sense. If this reading is right, the unresolved zone is the failure window for every behavior without a trained uncertainty-default, and refusal-style safeguards may be the well-covered exception rather than the rule.

Why the safeguard held: three candidate accounts. Three accounts could explain why the covered fiction/real task mostly safe-completed in the middle reading band, and we cannot yet separate them. They are not exclusive: one is about what training installed, one about what deeper layers see, one about where in the stack the trigger reads. None is causally established. With two tasks these are observations. The exploratory per-layer curves of §3.5 cannot separate them either. Shallow readings saturate after the shift, which blinds that instrument in exactly the layers the third account below points to.

A trained default. The first account is the reading above: the fiction/real task carries a trained default for unresolved cases, and the tank task does not.

Resolution at depth. The second account emerged from the depth data, and it is weaker than it first looked. In the fiction/real task, layers 3 to 18 settle clearly on the destination side in both directions, where the tank task from aquarium to vehicle ends at the midpoint (§3.2). Downstream layers in the fiction/real task may therefore act on a more settled reading than the site we measure. But the deepest layer band hovers near the midpoint in the fictional-to-real direction too, the direction in which safeguard behavior must appear, and did. Resolution at depth cannot carry the explanation alone.

An early, surface-keyed trigger. The third account points the opposite way through the stack. The shallowest layers resolve the framing composition almost immediately and completely (§3.2), with the profile of surface-cue tracking. The minimal-pair test that rules out cue-only tracking was run at the calibrated site, not at shallow layers. A safeguard that reads early, from surface content and frame cues, would fire whenever the alarming request is present and be suppressed only by a well-established fictional frame. That is the shape of our behavioral data, including its fragile 50% fiction-side floor. The fast, complete shallow response is specific to the fiction/real task. The tank task’s shallow layers respond later and, in one direction, only partway (§3.2). That is what a trigger sculpted by safety post-training would look like. But ordinary register statistics learned in pretraining predict the same asymmetry, and training provenance is unobservable here. Patching shallow against mid-stack states during generation is the decisive test (§5).

One limit. We state it so that this section cannot overpromise. As a standalone monitor, predicting each response from its reading alone, the frame reading is not yet usable: its AUC is 0.61 [0.43, 0.76], compatible with chance (Fig. 22). The band-level gradient of §3.5 is real. A usable monitor would combine several sites, the mixed-context marker, and per-layer readings. Building one is future work, not a claim.

A signal behavior does not appear to use. The mixed-context marker gives the sharpest form of the dissociation between what the model represents and what it does. The model carries a persistent, systematic signal that its context is mixed, and its behavior does not appear to use that signal when the mixture of frames in the context is held fixed, at the one site and layer tested. This is the structure reported in the hallucination literature, where models internally encode uncertainty or truthfulness that their generations do not respect [CITE: internal encoding versus expression]. Here it appears in transition dynamics. An internal signal correlated with the conditions that produce the unresolved zone already exists inside the model, and behavior does not appear to use it. Whether it could actually flag the zone is untested; the one monitor we built, from the frame reading, is compatible with chance. The motivating case of §1 reportedly had the same structure at system scale: the provider’s moderation layer was flagging the user’s messages for self-harm risk in real time while assistance continued [CITE: filing/reporting]. That suggests a constructive direction for alignment work: test whether behavior can be gated on the uncertainty signals already there, before building new ones.

The human parallel. The §3.2 signature is a partial reanalysis that leaves a lingering trace of the initial misreading. That is also the signature of human sentence processing in the “good-enough” tradition [CITE: good-enough processing]. After recovering from a garden-path sentence, comprehenders measurably retain components of the initial, incorrect interpretation. We make no mechanistic identification. We note that a language model trained on human text reproduces, at the activation level, the reanalysis profile humans show behaviorally, including the part where the old reading never fully leaves.

Garden path or pun. In the terms of Dynel’s account of garden-path humor [CITE: Dynel], our transition corpus is a slow-motion garden path, and most trajectories treat it as one: incongruity, then resolution toward the new reading. The dwelling runs are the interesting case. They form an extended incongruity phase, in which the model, asked what the word means, answers like someone explaining a pun: both senses, held. Whether any of them is a true pun, a held incongruity that never resolves, or only a long garden path is the persistent-versus-slow question our twenty-sentence horizon cannot decide. The extended-tail experiment named in §5 is its dedicated test.

Closing. Word-sense reinterpretation is the tractable laboratory instance of a much larger family. Frames, personas, tasks, and safety postures all shift under accumulating context, and there is no obvious reason the metastable structure documented here is unique to word senses. The states are what the title calls them: unresolved. For now, so is the question of what a model should do while inside one.

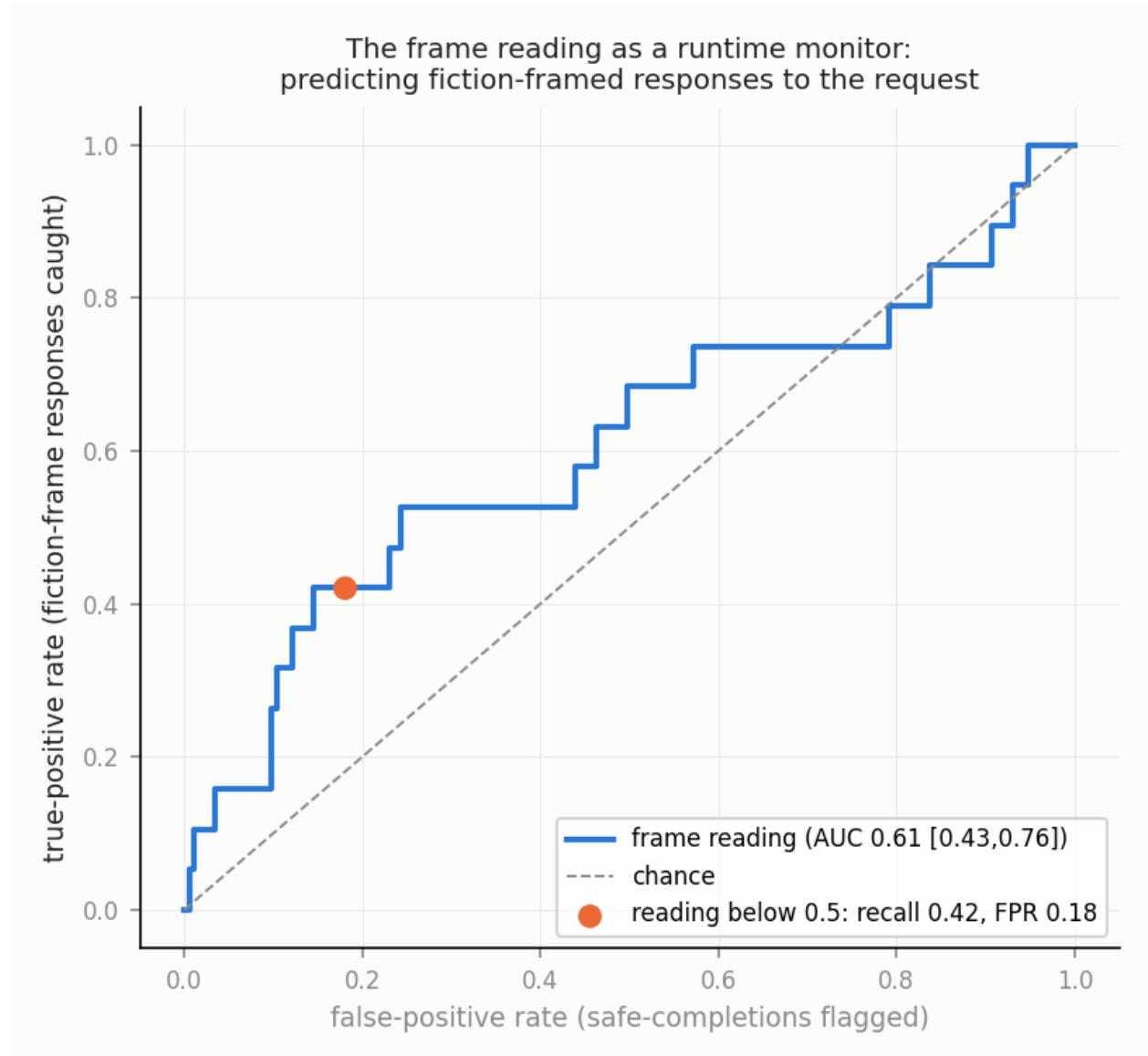


Figure 22: The frame reading as a standalone monitor, an exploratory check added after the analysis freeze. Receiver operating characteristic for predicting fiction-framed responses to the request (responses that take up the fictional frame rather than safe-completing) from the calibrated-site reading alone: the true-positive rate (fiction-framed responses caught) against the false-positive rate (safe-completions flagged). The AUC is 0.61 [0.43, 0.76], with a 95% interval clustered by scene family (§2.3), compatible with chance (dashed line). The marked point is one threshold, with its recall and false-positive rate.

5 Limitations and future work

One model, stated plainly. Everything here is measured in one 20-billion-parameter mixture-of-experts model, with deterministic decoding. Each dynamics claim is made at one calibrated site and layer per task — single rows of a depth progression in which crossing timescales vary systematically (§3.2); no single layer is a sufficient readout of the model’s state, and behavior is produced downstream of all of them. Within the study, replication is internal: two task contrasts, three carriers plus a replicate, two measurement sites, twelve scene families per class per task. That inventory also records what did *not* replicate cleanly: the real→fictional remnant is suggestive only once reference uncertainty is propagated; the letter-site asymmetry rests on four runs per direction; 48% of fiction/real runs are indeterminate under per-run model selection; and our pre-stated prediction about the ordering of crossing times across layers held in one task and failed in the other. Cross-task patterns — the task with the wider endpoint separation also shows the larger remnant gaps; anchored versus distributed contrast structure — are $n = 2$ observations, hypotheses rather than findings.

The named open question. Whether the remnant — and the dwelling within the unresolved zone — are permanent or merely slower than twenty sentences: in the terms of §4, whether the dwelling is a pun rather than a long garden path — is decidable by extending the post-shift horizon (extended-tail runs) and by shifting evidence back to the original class ($A \rightarrow B \rightarrow A$ runs, measuring what remains of the second frame). Both are designed and costed; neither has run.

Further deferred work, in rough order of leverage:

- Replicating the fiction/real headline quantities at the ‘write’ site, where the framing contrast reads most strongly (the existing recordings suffice).
- Expanding the letter-site families.
- Calibration sets from a third, unrelated class, to identify what the accumulation offset contains.
- Monitor construction (multi-site, marker-augmented, depth-aware) against the chance-compatible single-site baseline reported above.
- Causal localization of where safeguard behavior reads in the stack: patching shallow versus mid-stack states during generation — the decisive test for §4’s three candidate accounts.
- A planned suite of harder context manipulations — colliding frames, mutating them mid-context, deliberately incoherent contexts — where genuine off-distribution excursions are most likely, our clean block-shifts being the tamest possible case.

The analyses reported here were frozen before drafting; the post-freeze additions are logged as such in Appendix B.

Back matter (assembly spec — materialized at final build)

Review-draft note: the sections below are the assembly specification; the final build replaces them with the referenced frozen content.

Acknowledgments

[Agreed AI-assistance line — exact text from Andrew/chat side, placeholder.] Personal acknowledgments: Andrew adds.

Related work (arXiv default: standalone section, drafted from handoff §6 anchors)

Probing validity: Alain & Bengio; Hewitt & Liang (control tasks — our D5 + label-shuffle audits presented as the selectivity answer); Belinkov survey. Directions & steering: Marks & Tegmark (also the mass-mean/diff-of-means probing precedent for §2.2’s axis choice); Ardit et al. (refusal direction — nearest cousin of the fiction/real behavior link). In-context dynamics: Xie et al. (ICL as implicit Bayesian inference — the standing theory our integrator null speaks to); Hendel; Todd (function vectors); Olsson et al. (induction heads). Safety: many-shot jailbreaking; Crescendo-style multi-turn escalation. Psycholinguistics: Christianson & Ferreira; Slattey et al.; van Schijndel & Linzen. Verbalized uncertainty & ambiguity (grounds the intro’s scoped claim): Kadavath et al.; Lin et al. (verbalized confidence); Xiong et al. (calibration of verbalized confidence); Liu et al. (ambiguity modeling). Internal encoding vs expression (grounds Discussion’s “represented but not consulted”): Azaria & Mitchell (internal state predicts truthfulness); Orgad et al. (models encode more than they express). Long-context artifacts: attention sinks (the “is the offset positional?” question; midpoint referencing makes our findings robust either way). Dynamics: Sussillo & Barak; Kelso; Rabinovich et al. Motivating case (intro + discussion): Raine v. OpenAI, Cal. Super. Ct. (S.F.), filed 2025-08-26; contemporaneous reporting (CNN 2025-08-26; NBC News; TechCrunch 2025-11-26 on OpenAI’s answer denying causation). Contested litigation — the paper characterizes only what the filing and reporting describe. Model provenance (the intro’s “trained safeguard”): gpt-oss model card, safety training documented.

Appendix A — Corrections affecting reported values

One line each for the corrections that changed numbers as printed: the real→fictional remnant interval widening to include zero under reference resampling; the trimmed within-stream means; the per-run crossing-time medians; the recency-parameter memory figures. The complete corrections record is in the repository.

Appendix B — QA and reproducibility

Regeneration audit summary (bit-identical axes; 23-script diff green; seeded bootstraps exact); 9/9 fixture suite; sign/boundary/join/shuffle audits; the four logged post-freeze additions (s12 response counts, 0/96; s11 monitor ROC, AUC 0.61 [0.43,0.76]; s13 per-layer collapse panels; s14 exploratory per-layer behavior association — both computed from frozen captures via the committed cache); repository pointer + regeneration instructions. Terminology map for repository readers: the paper’s *remnant* is the repository’s “residual”; the dwelling within the unresolved zone is the repository’s “park”; *accumulation offset* is “accumulation drift”; *tasks* are the repository’s “probes”/“probe arms”; corpus directories D3/D4/D5/D6 are the transition, no-shift, minimal-pair, and mixture-sweep corpora; the repository’s “K=1” label names an analysis convention for a routing track this study dropped.

Appendix C — Supplementary figures

figures/other/ (12): fr fit gallery, fr D6 loop, tank secondary heatmap, matched-k panel is MAIN (`fig_s9_behavior_matchedk`) — supplement gets: raw + midref heatmap generations (the instrument story’s “before” pictures), `prelexical_L4`, `occupancy_bands_L4`, `jumpiness`, `norm_vs_alignment`, `calibration_layers`, plus `fig_s11_monitor_roc`. `fig_s13_collapse_layers` is MAIN (cited in §3.2’s depth paragraph; committed script `s13_collapse_by_layer.py`).